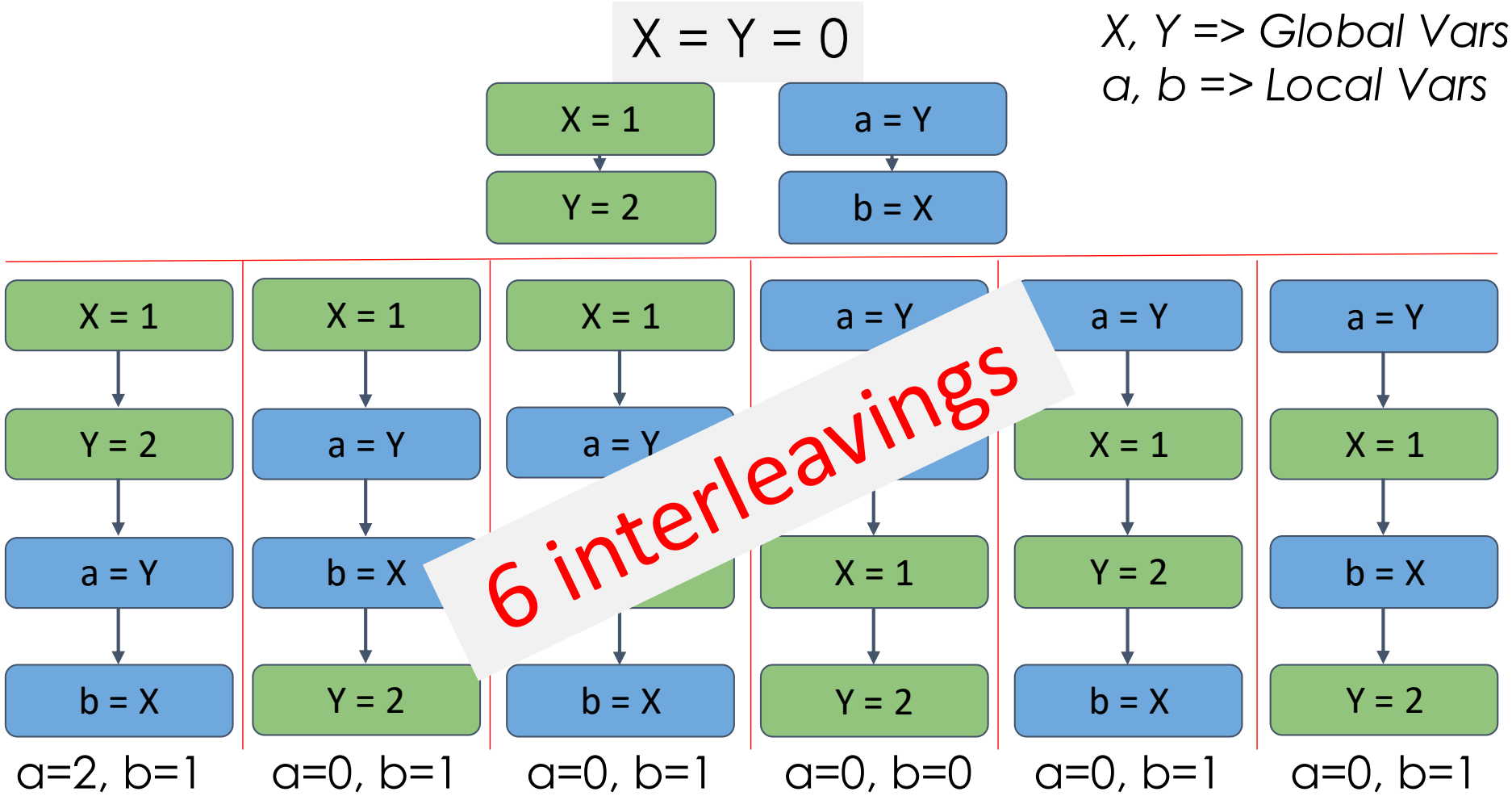
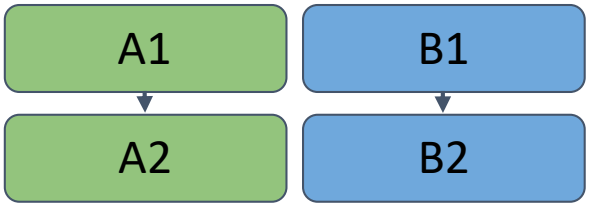
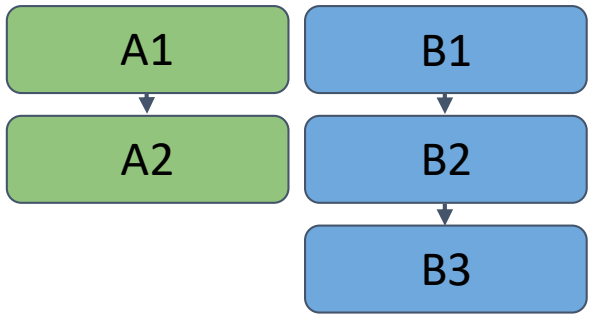
State Space Explosion



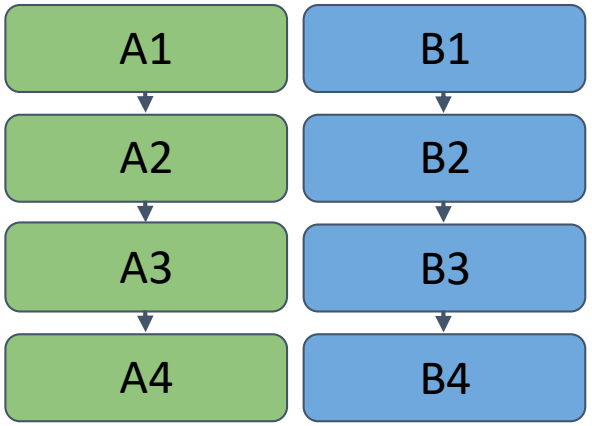
State Space Explosion



6 interleavings



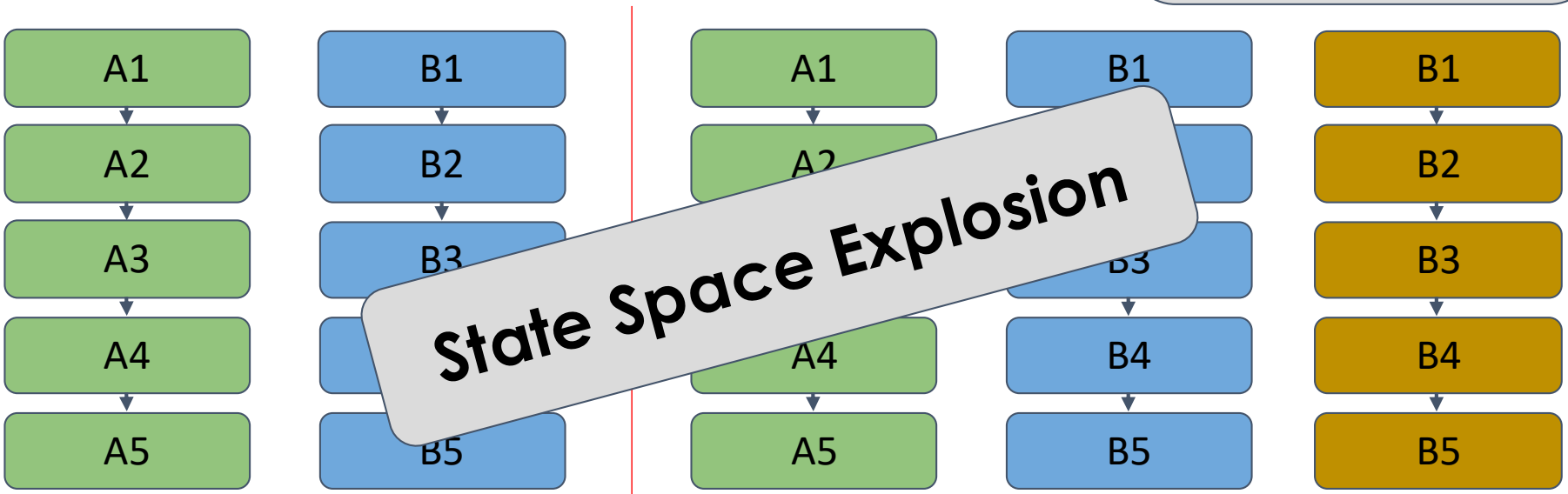
10 interleavings



70 interleavings

State Space Explosion

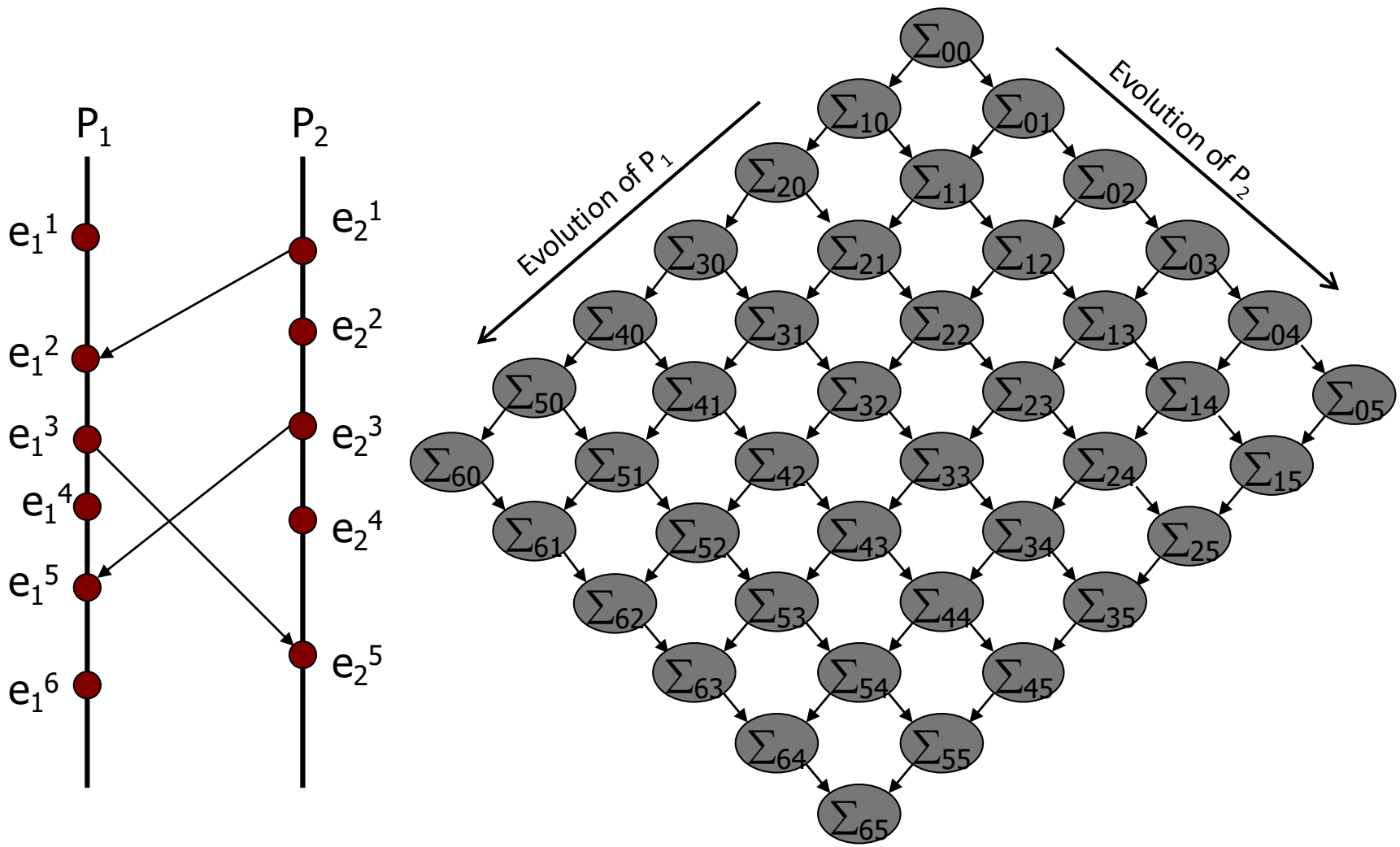
$$N = \frac{(\sum T_i)!}{\prod(Ti!)}$$



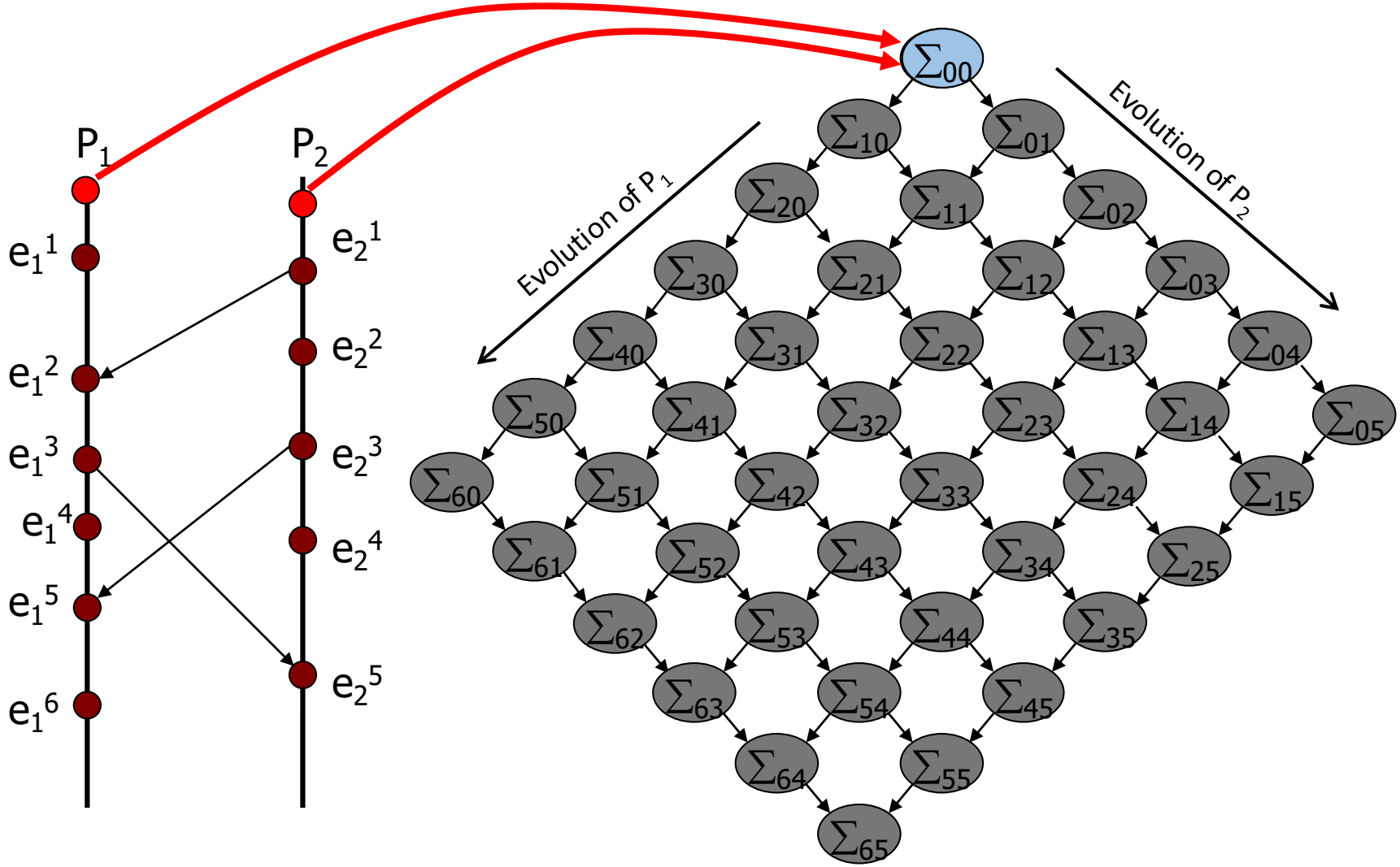
252 interleavings

756 756 interleavings

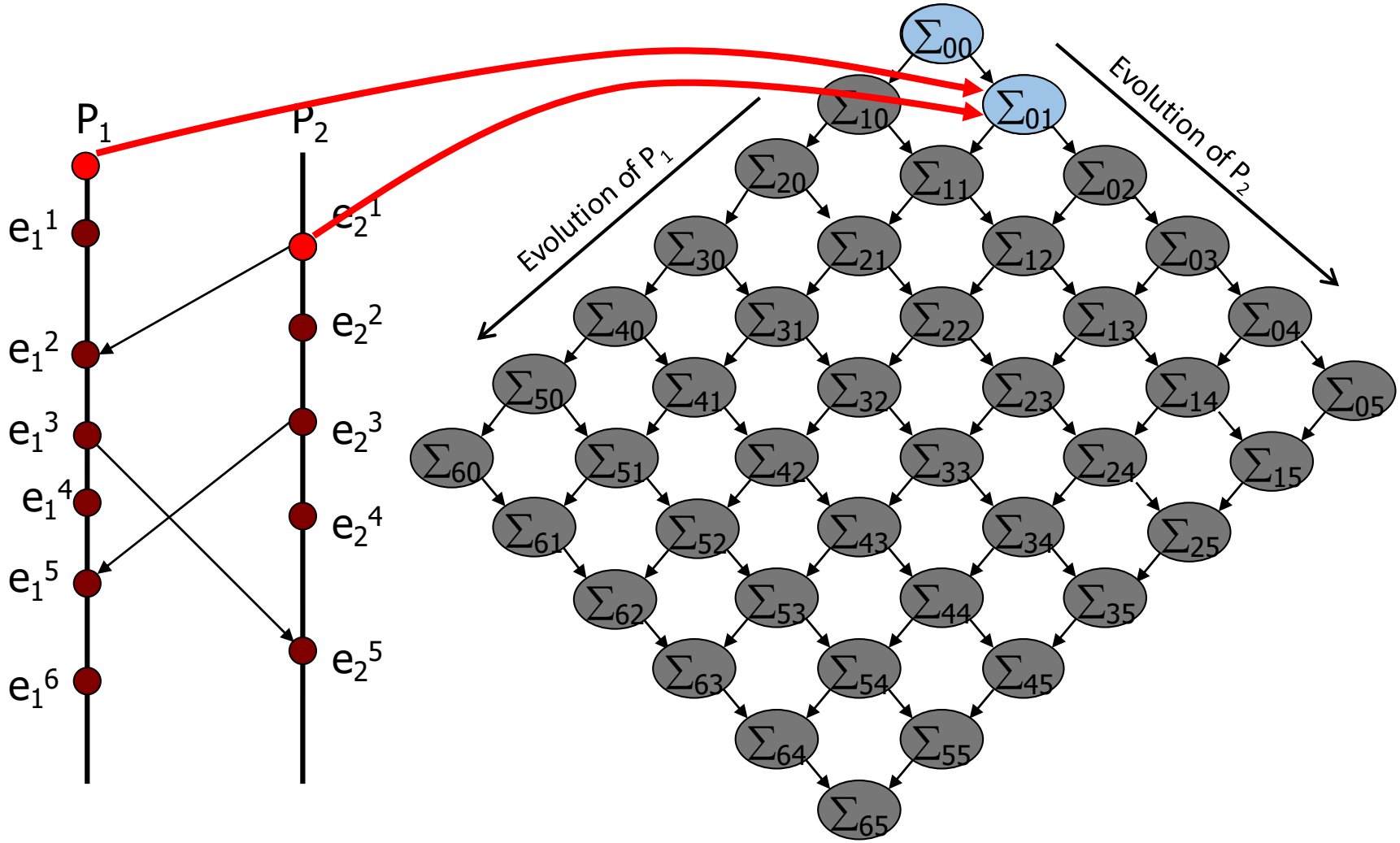
State explosion in concurrent programs



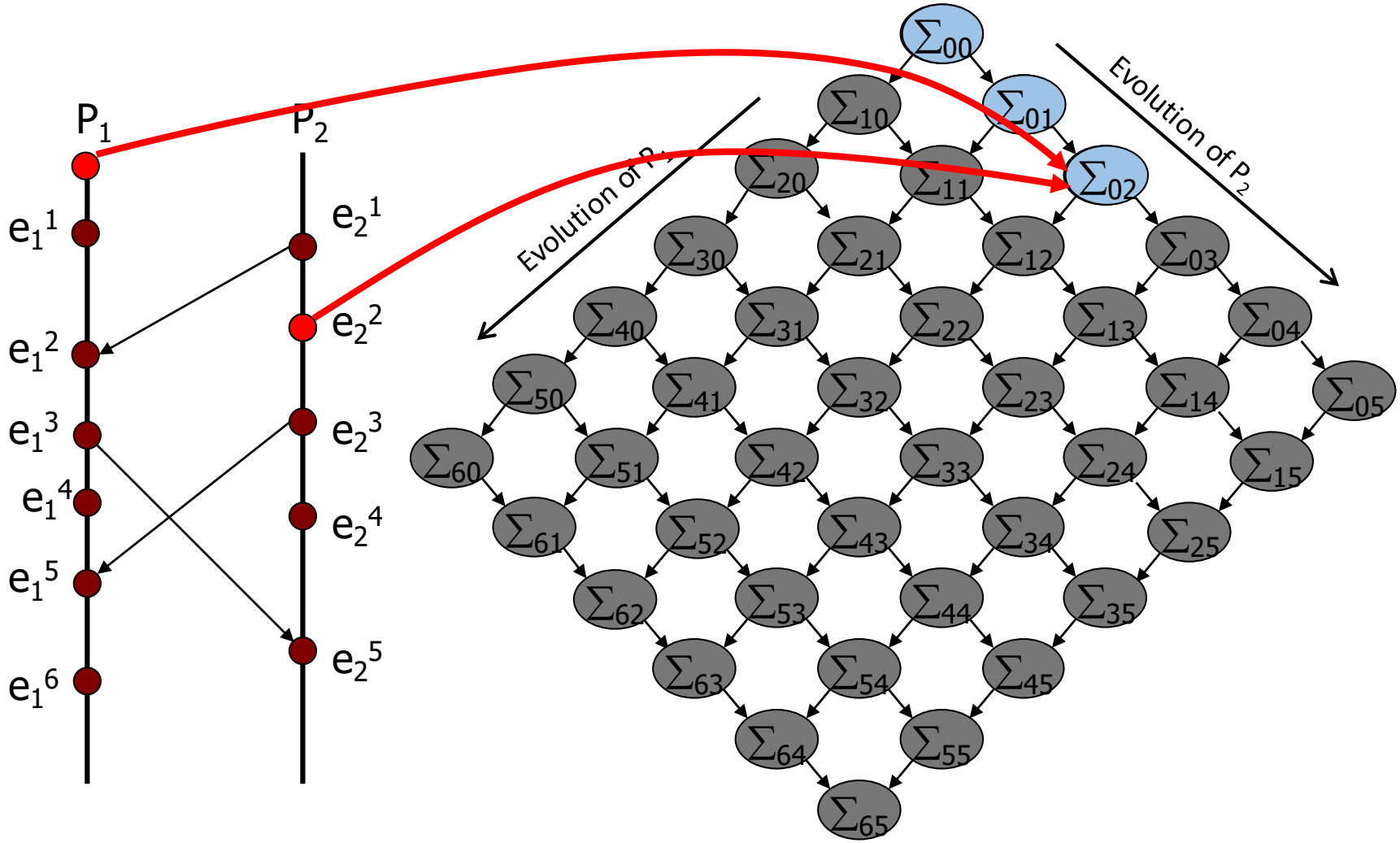
State explosion in concurrent programs



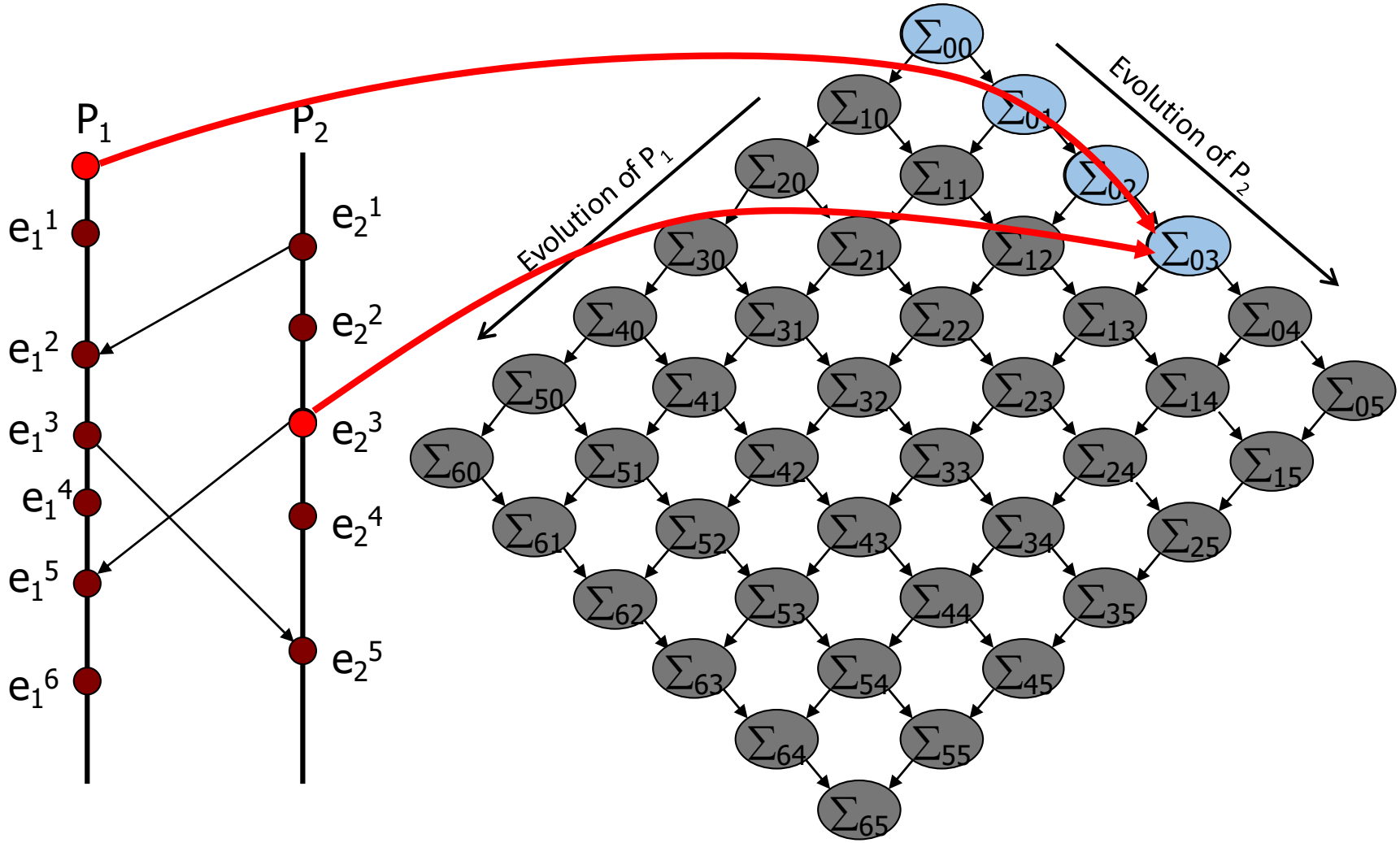
State explosion in concurrent programs



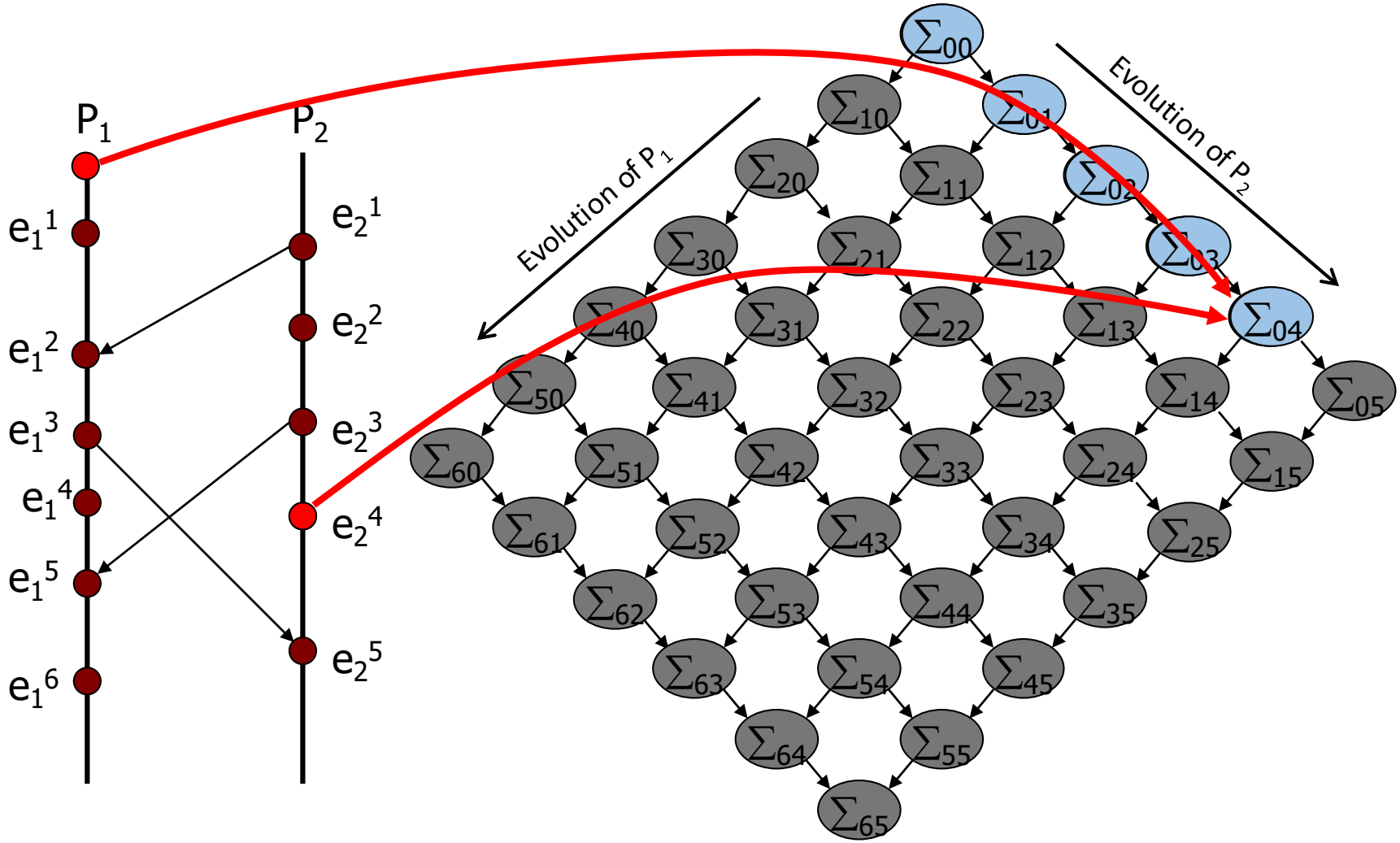
State explosion in concurrent programs



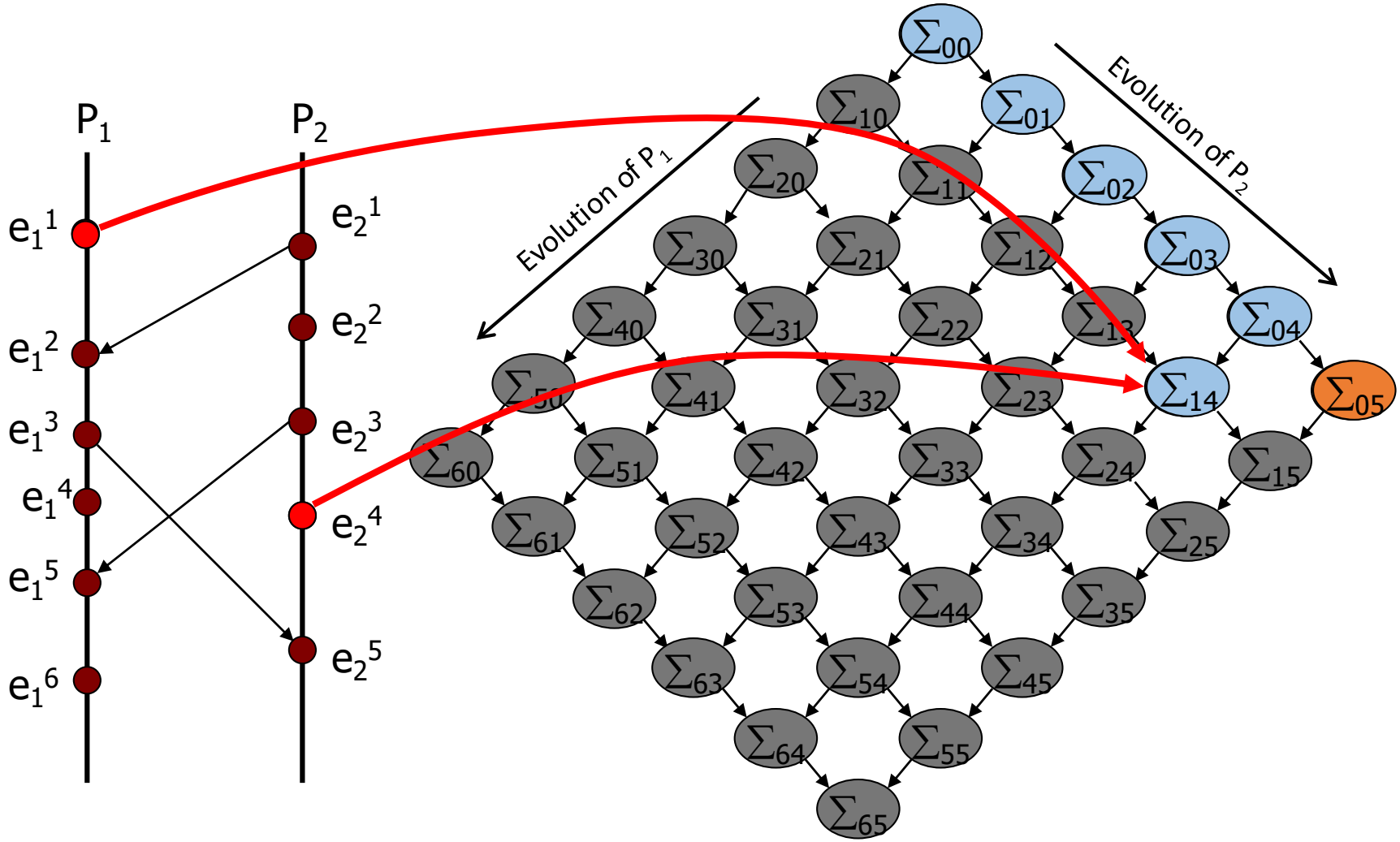
State explosion in concurrent programs



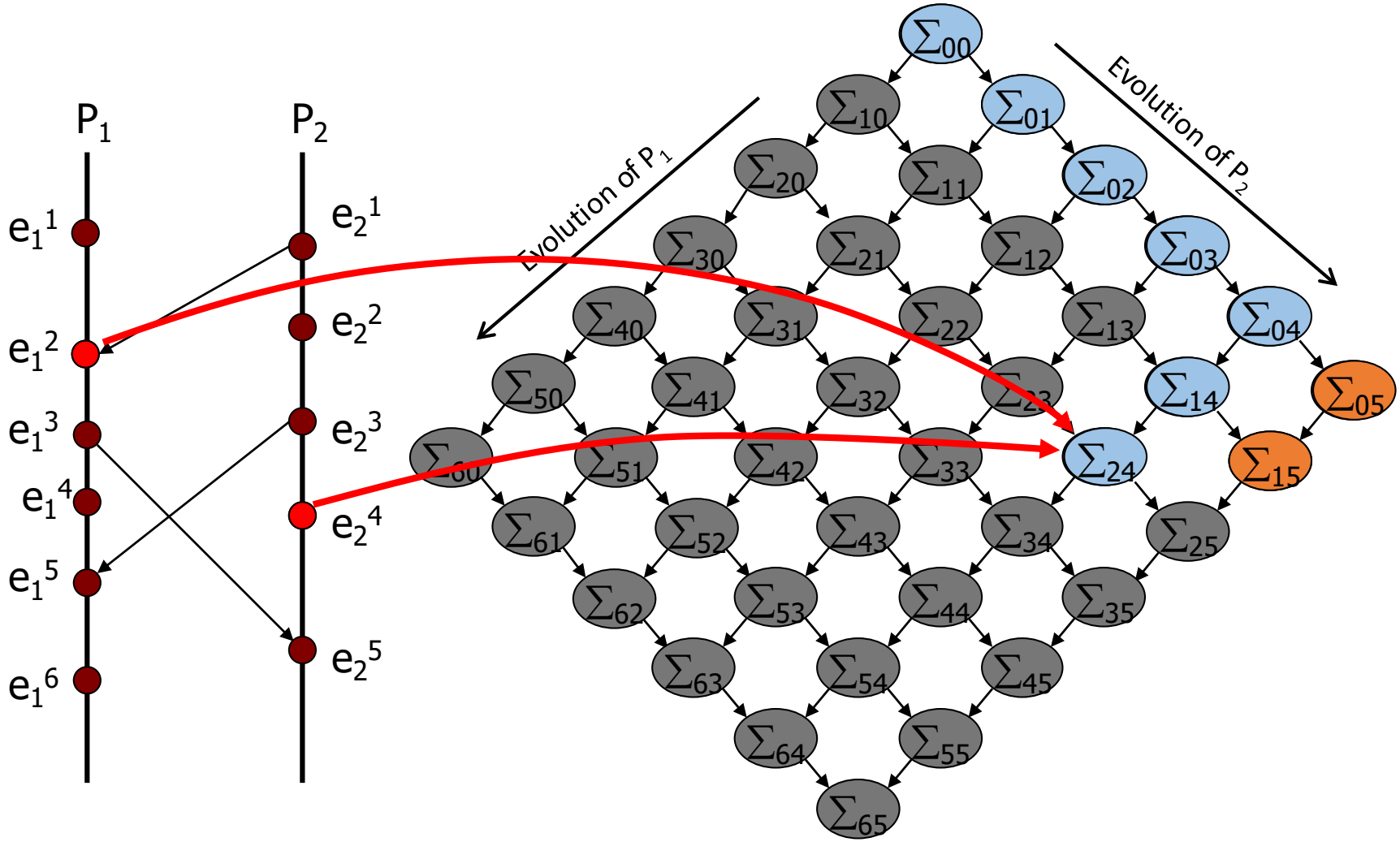
State explosion in concurrent programs



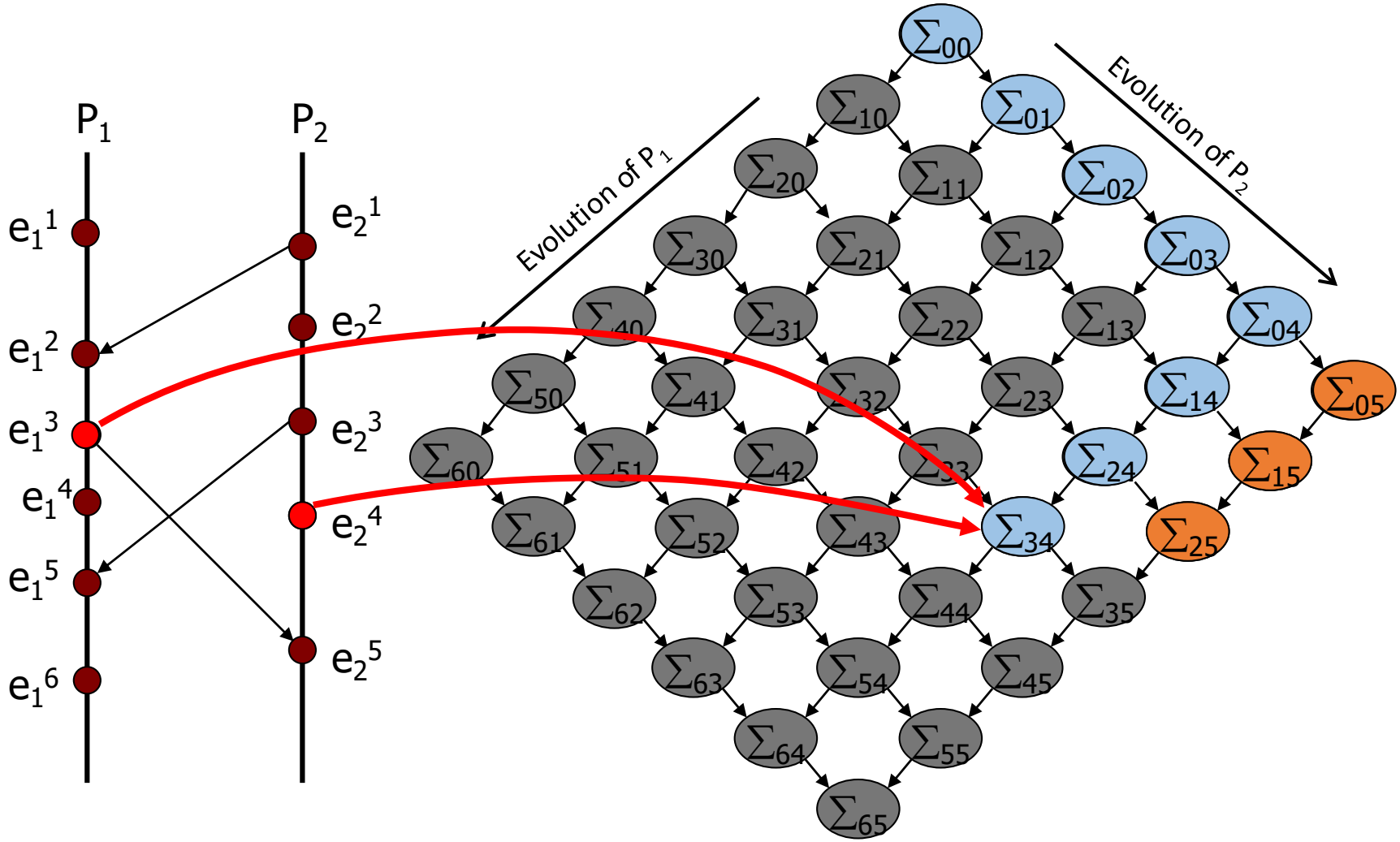
State explosion in concurrent programs



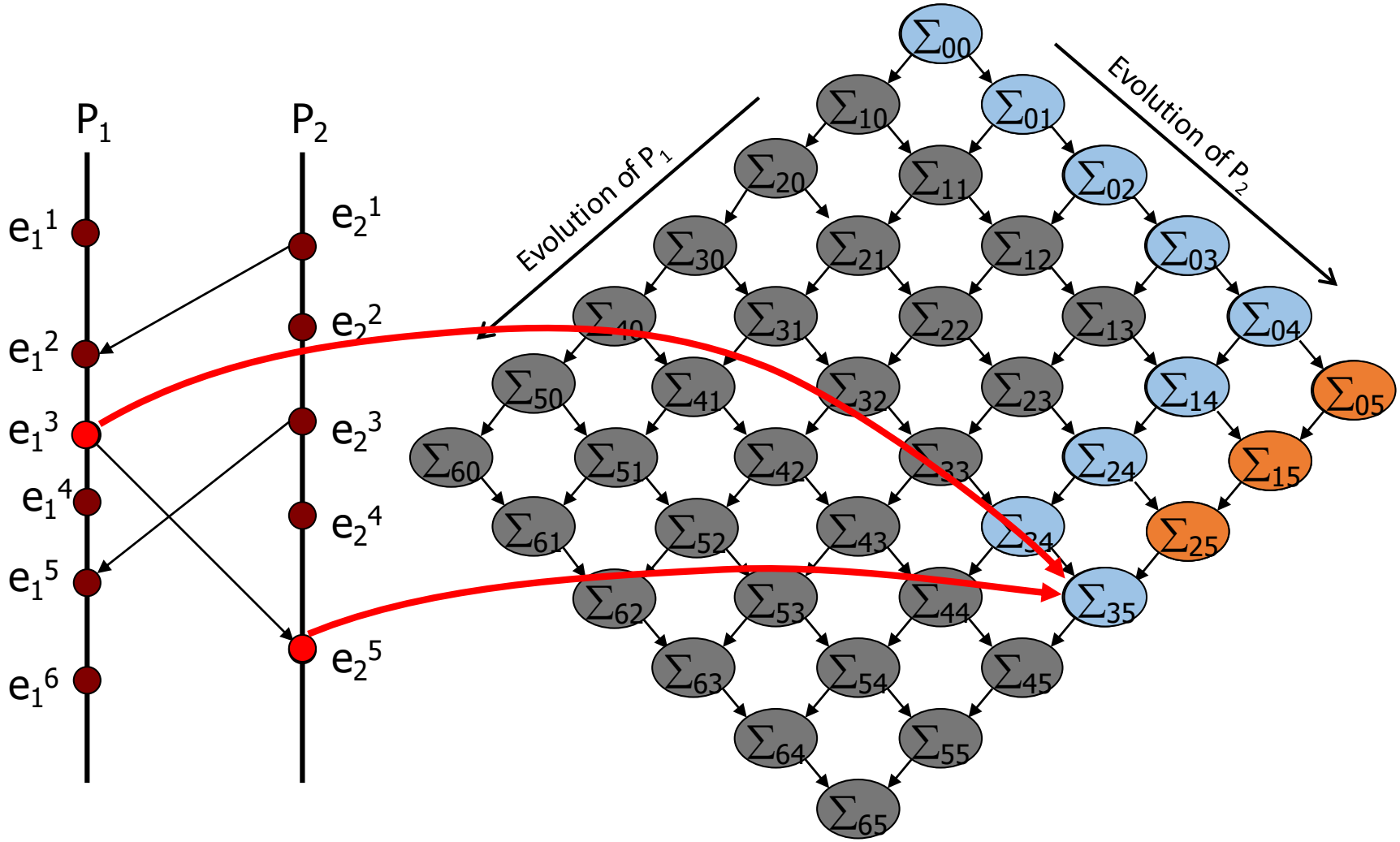
State explosion in concurrent programs



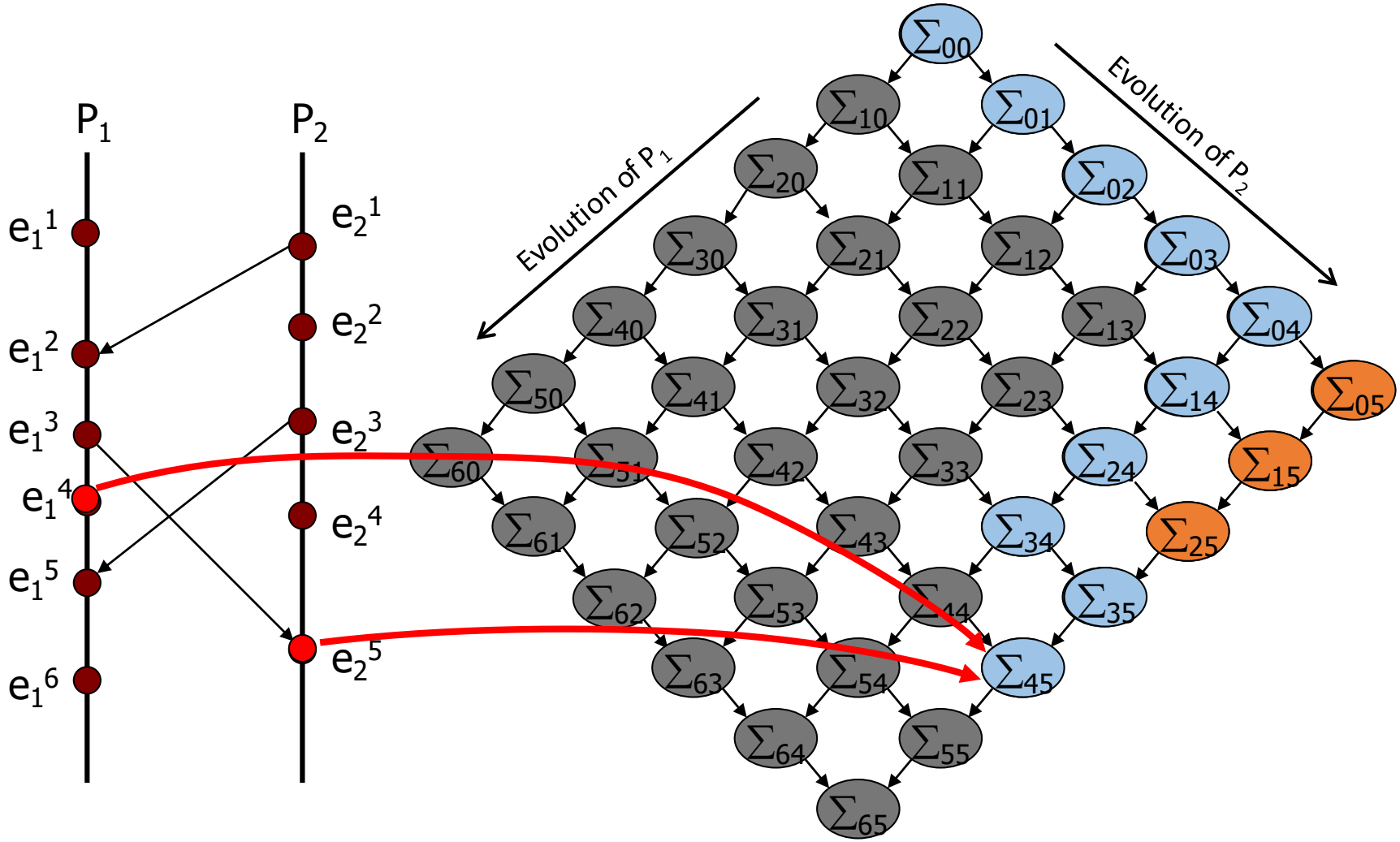
State explosion in concurrent programs



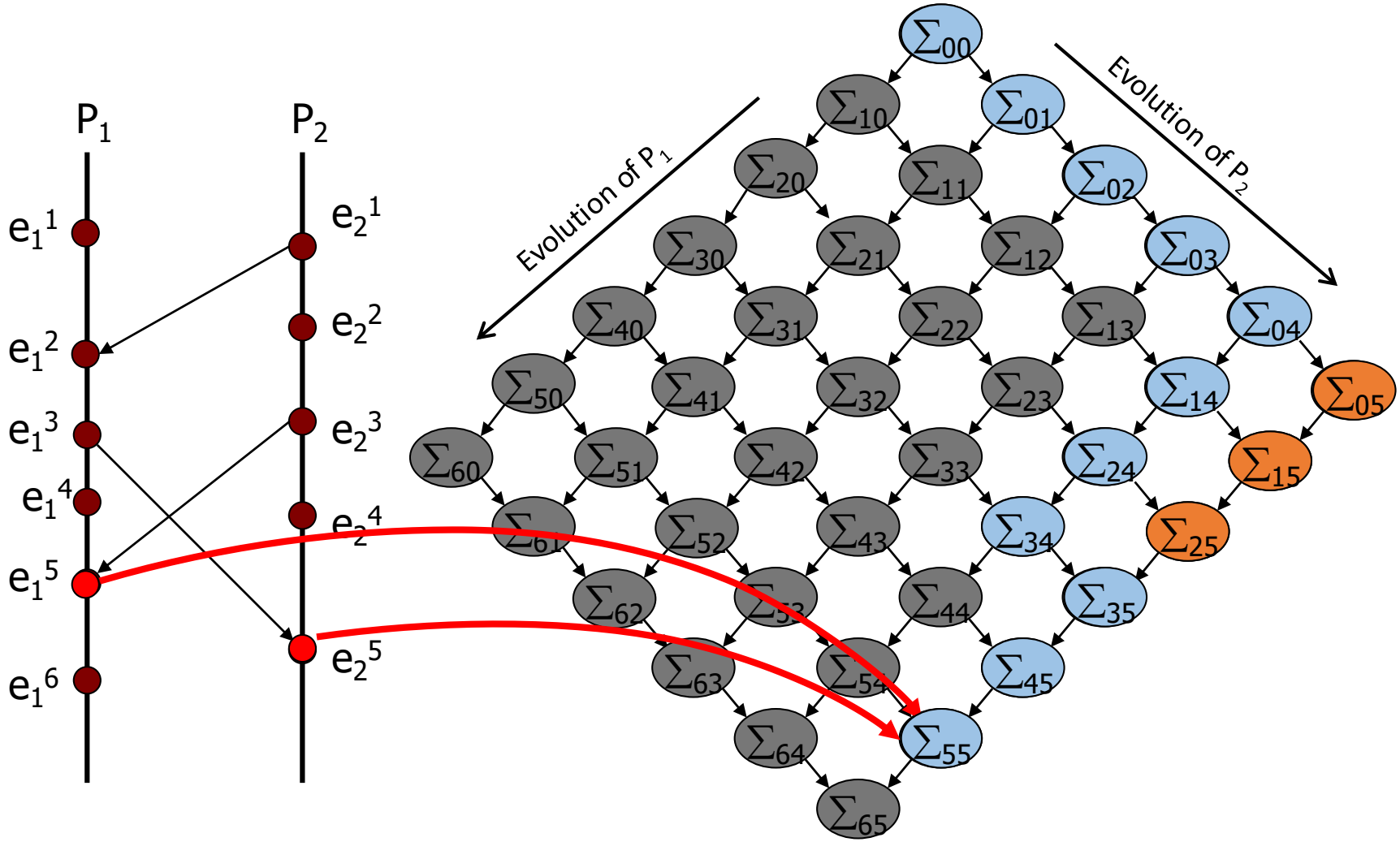
State explosion in concurrent programs



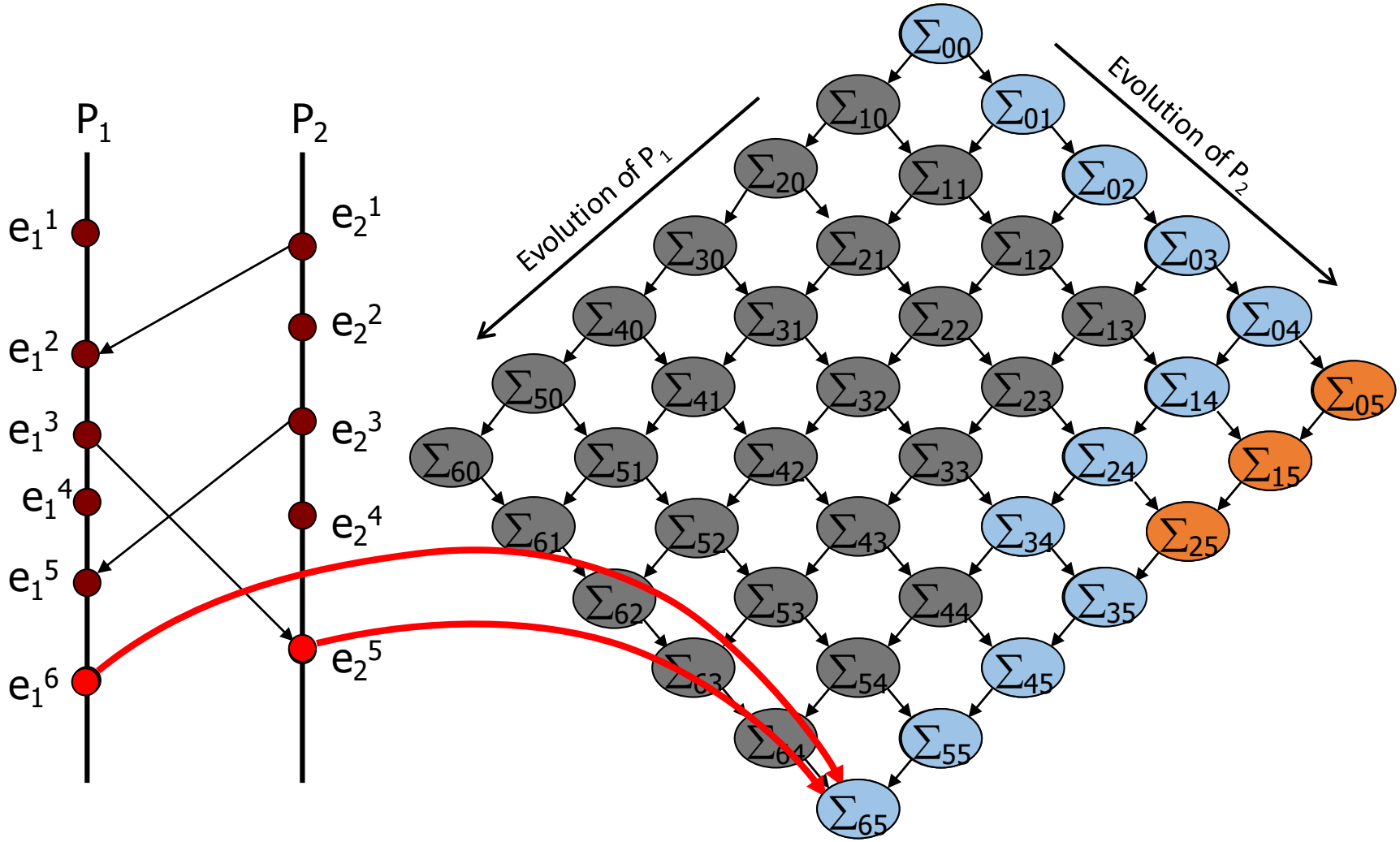
State explosion in concurrent programs



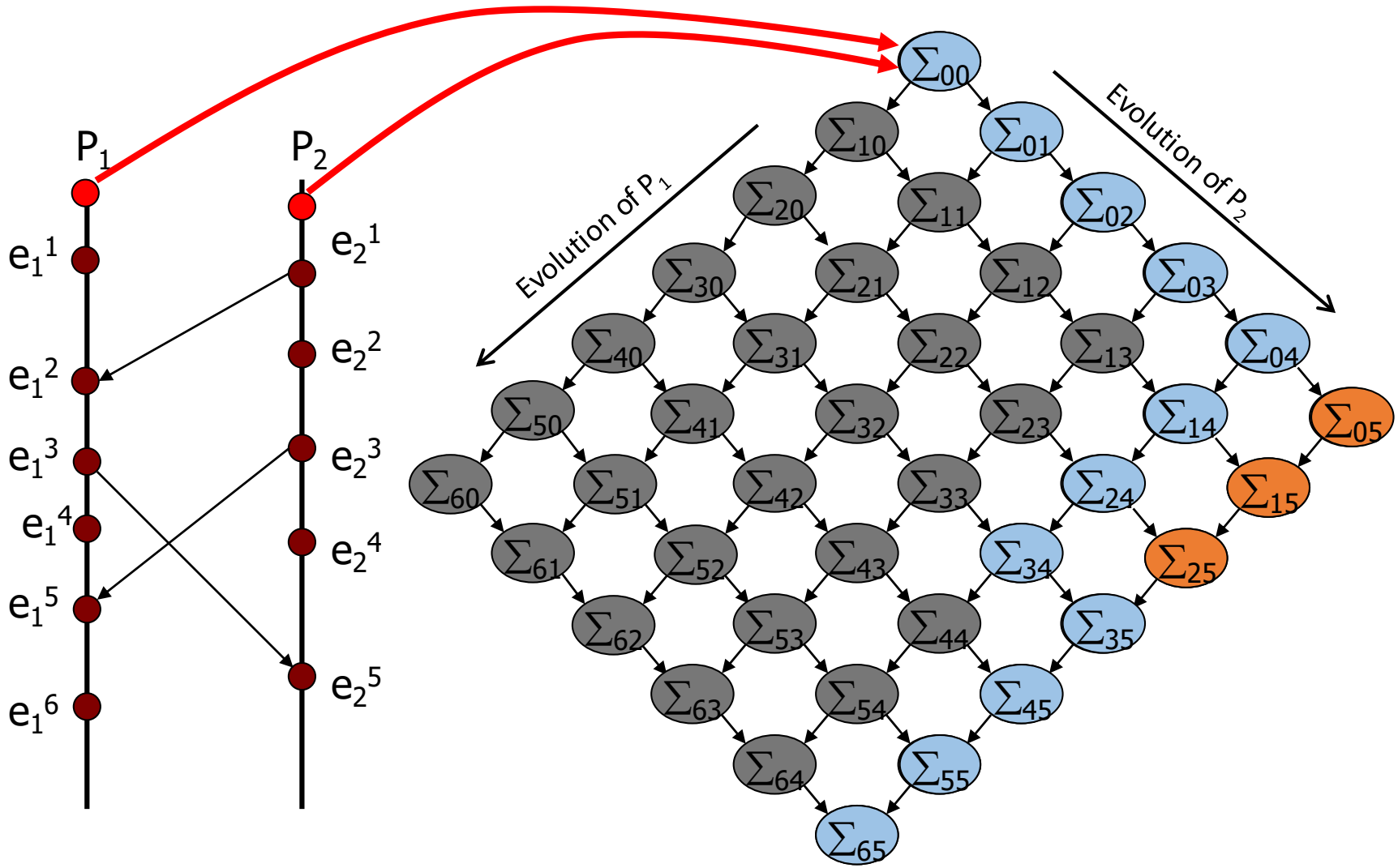
State explosion in concurrent programs



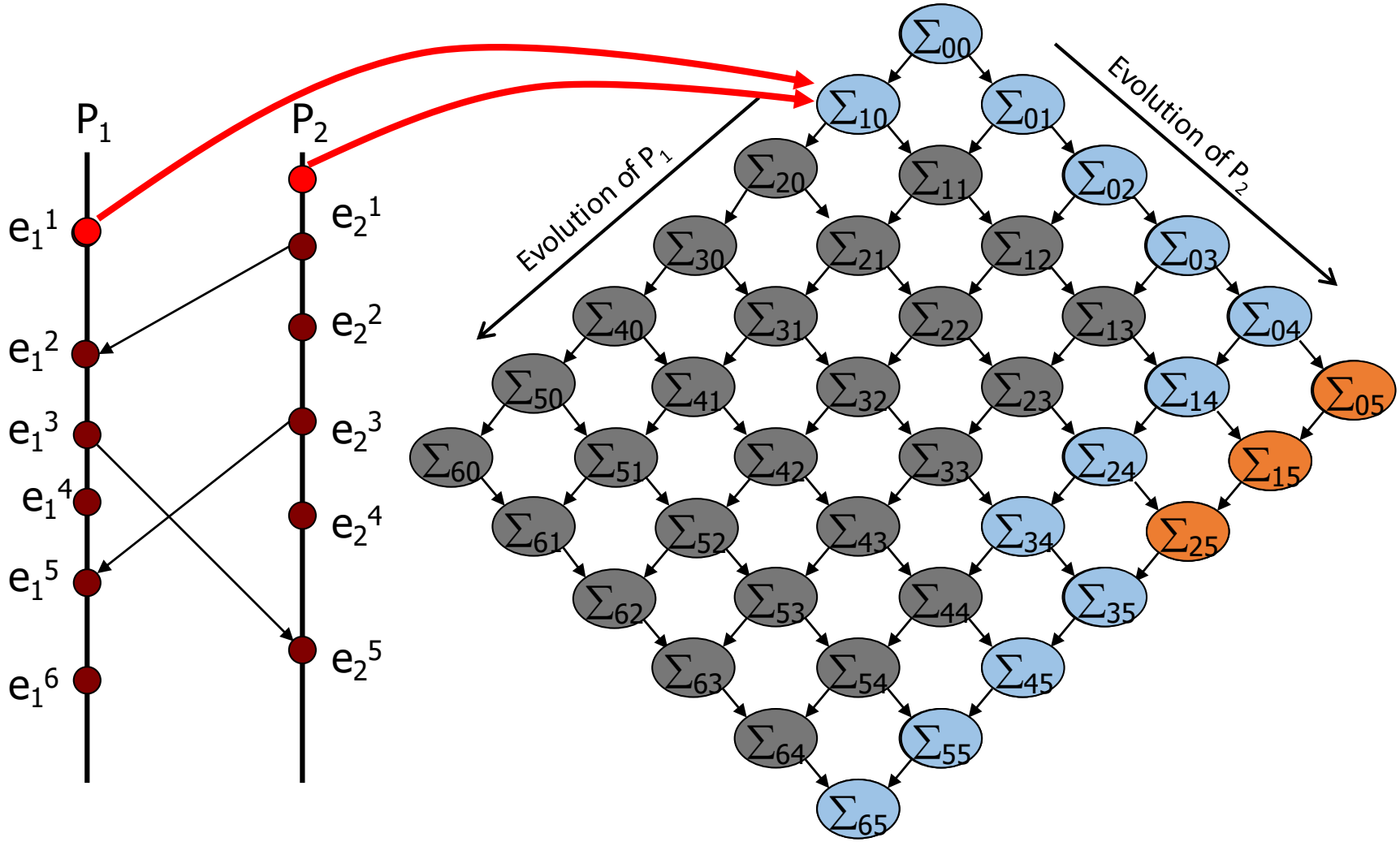
State explosion in concurrent programs



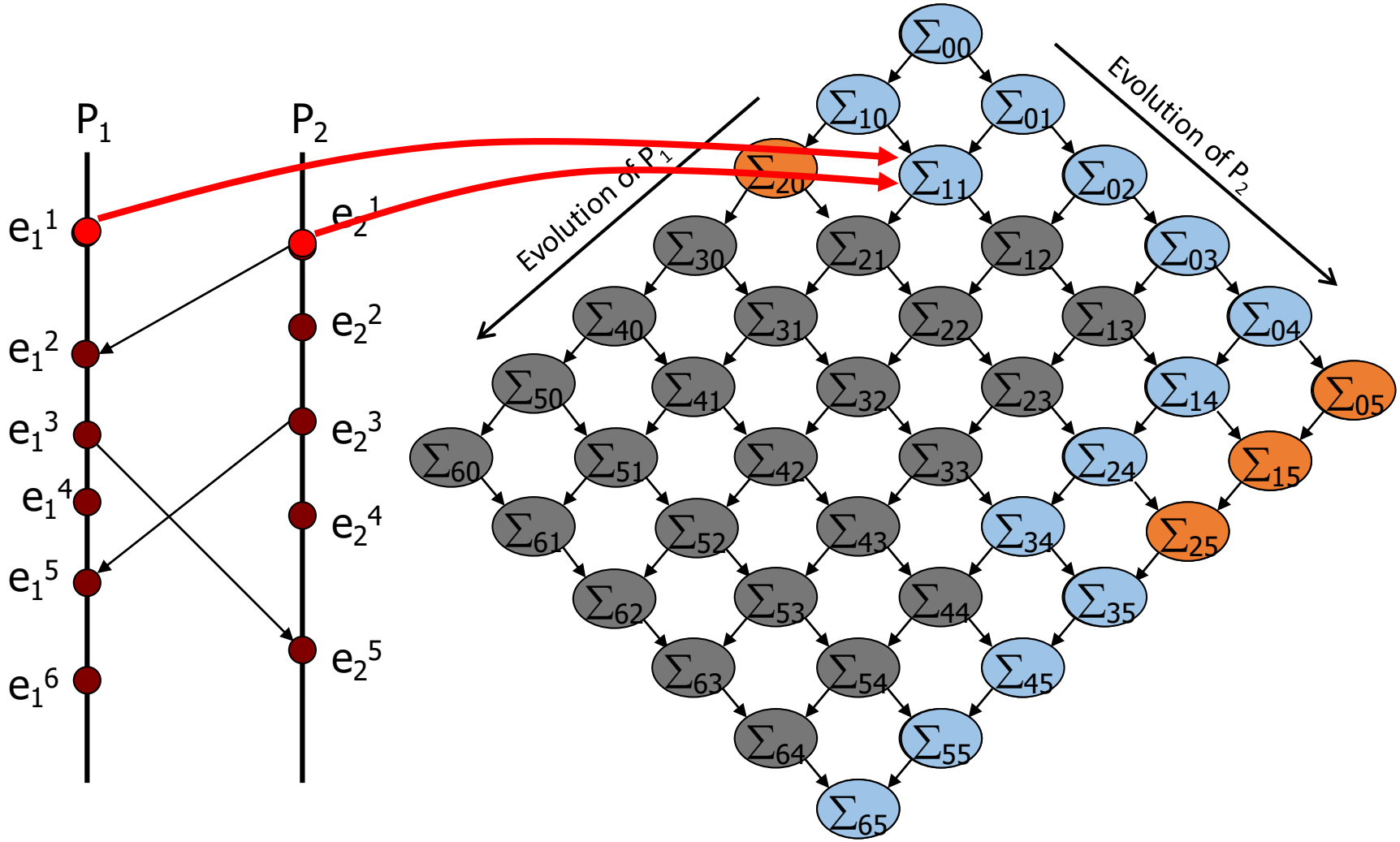
State explosion in concurrent programs



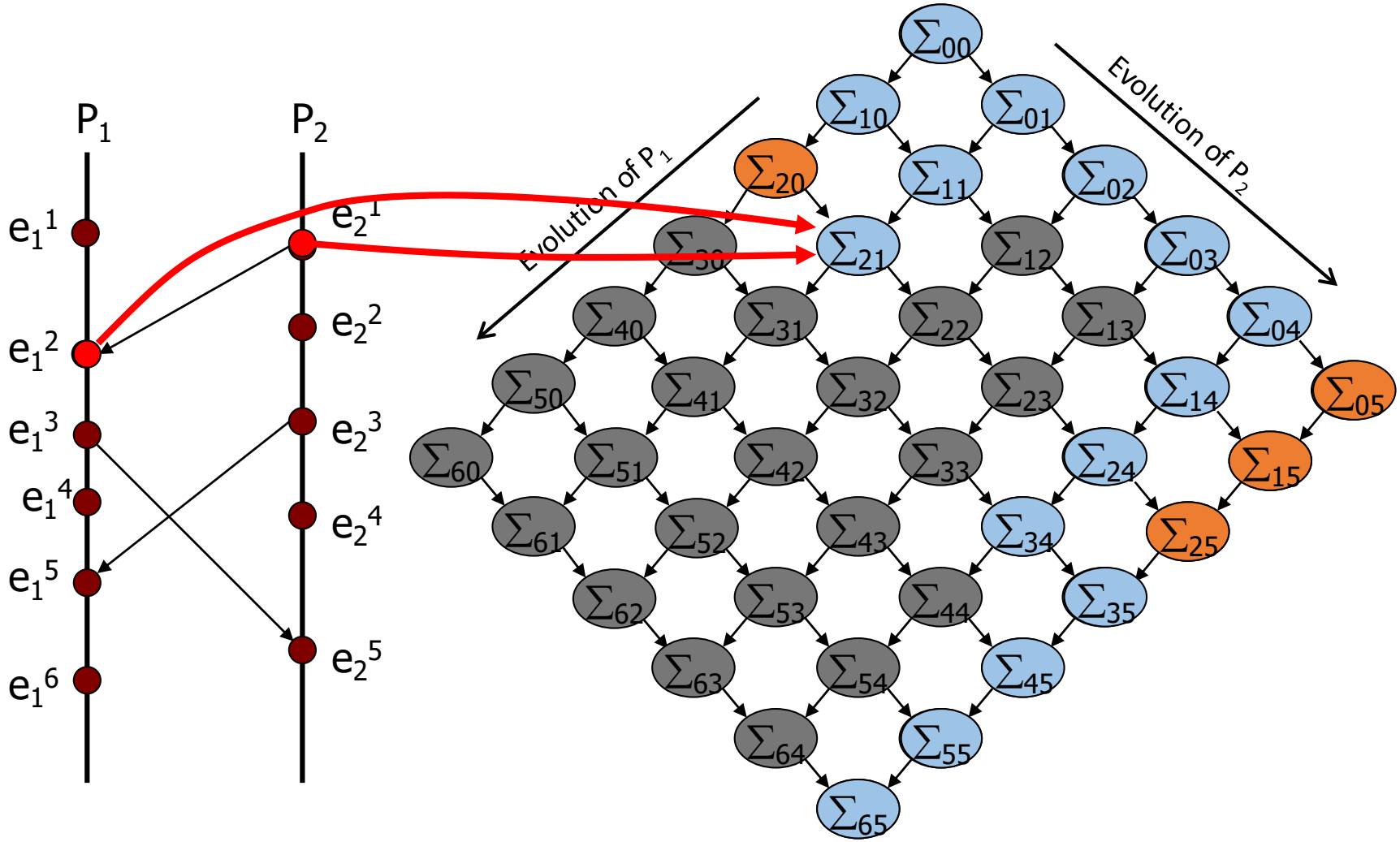
State explosion in concurrent programs



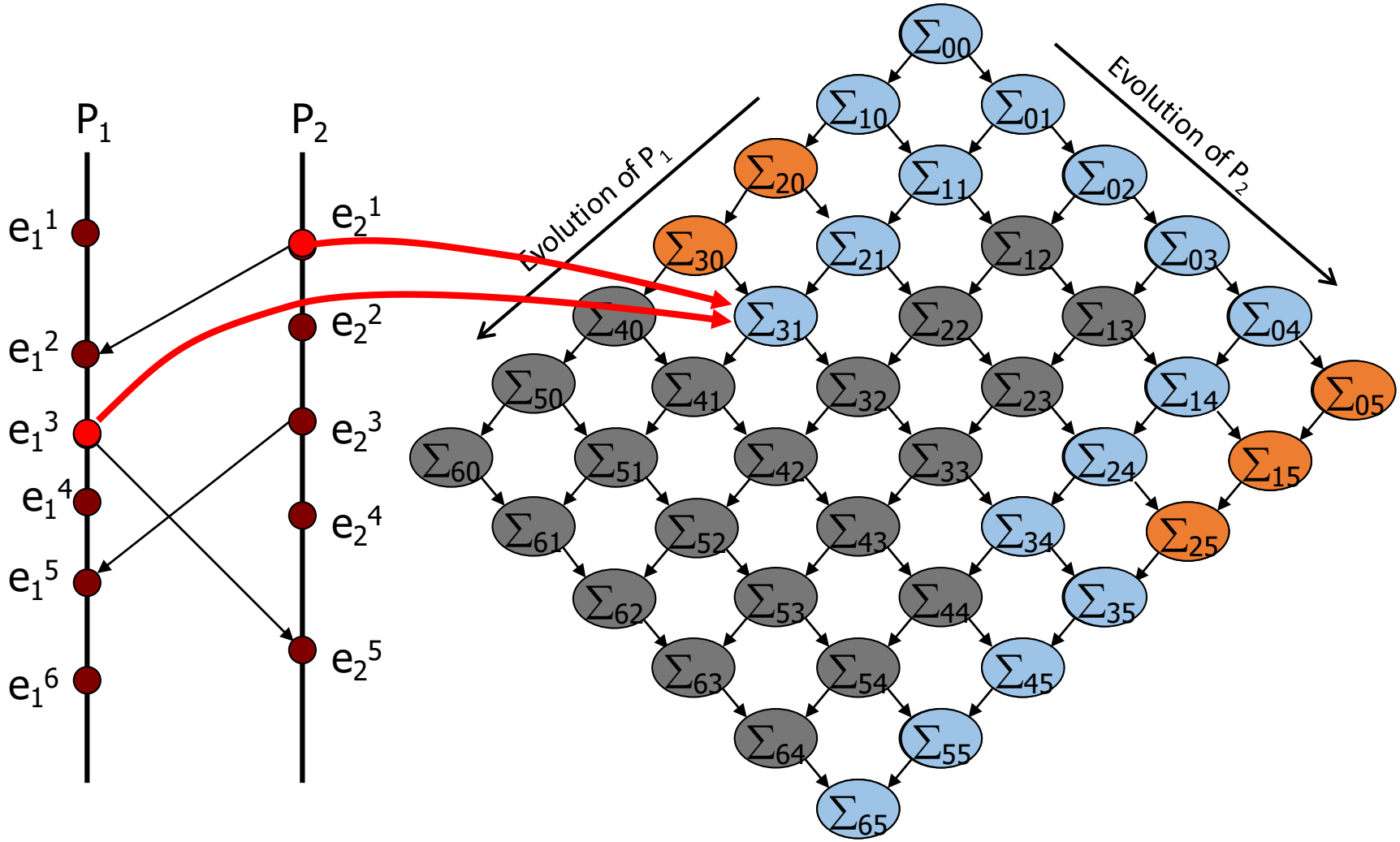
State explosion in concurrent programs



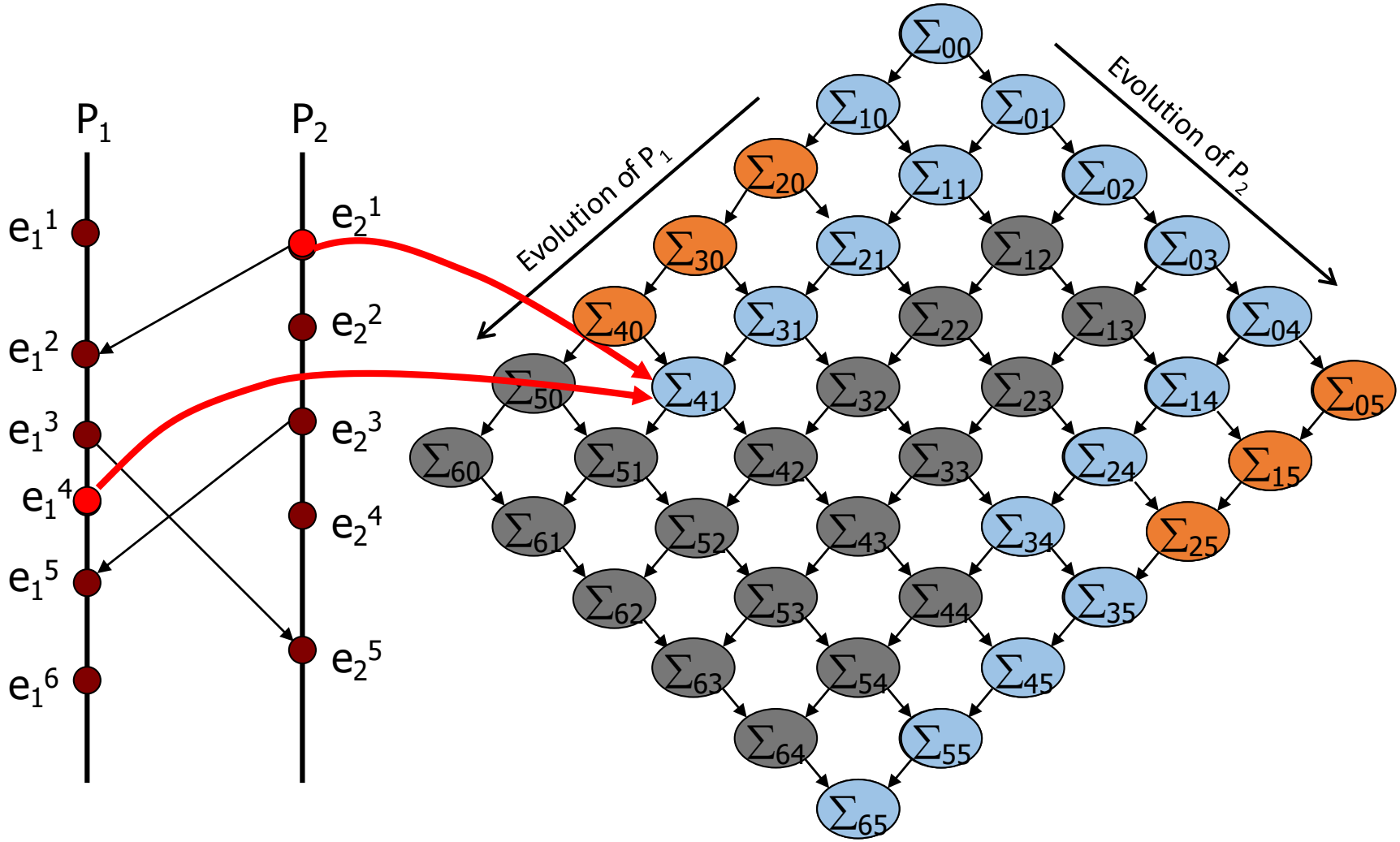
State explosion in concurrent programs



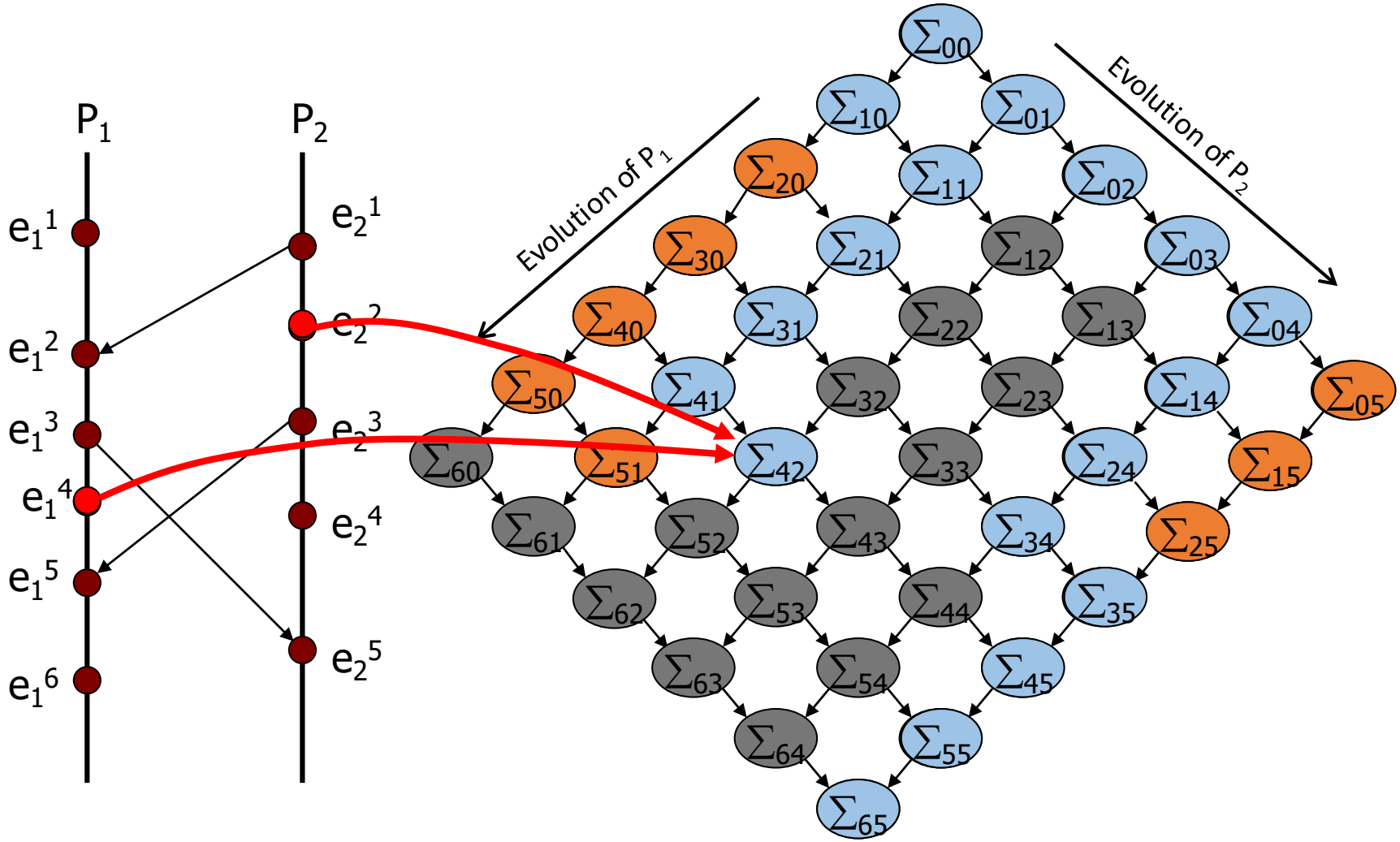
State explosion in concurrent programs



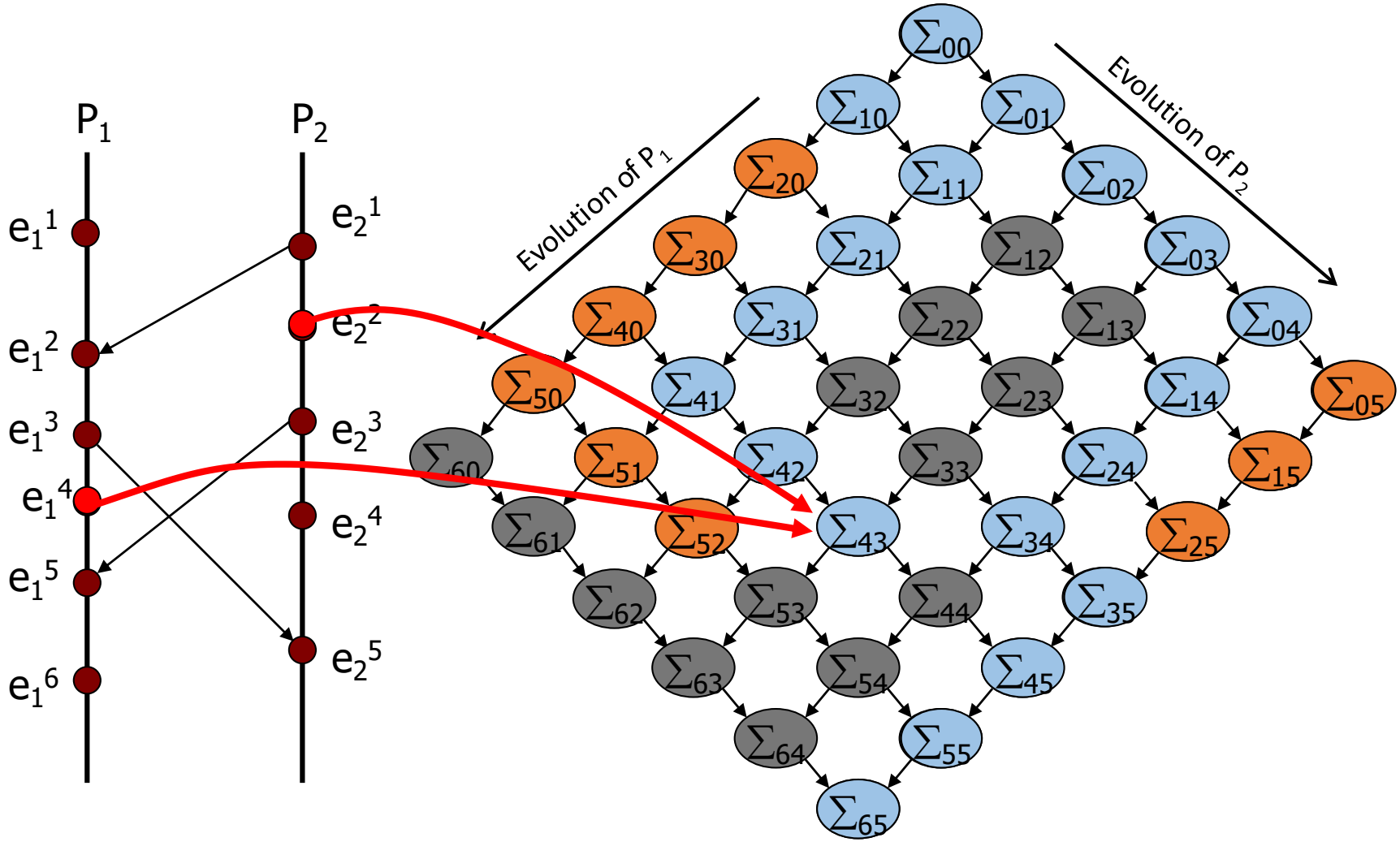
State explosion in concurrent programs



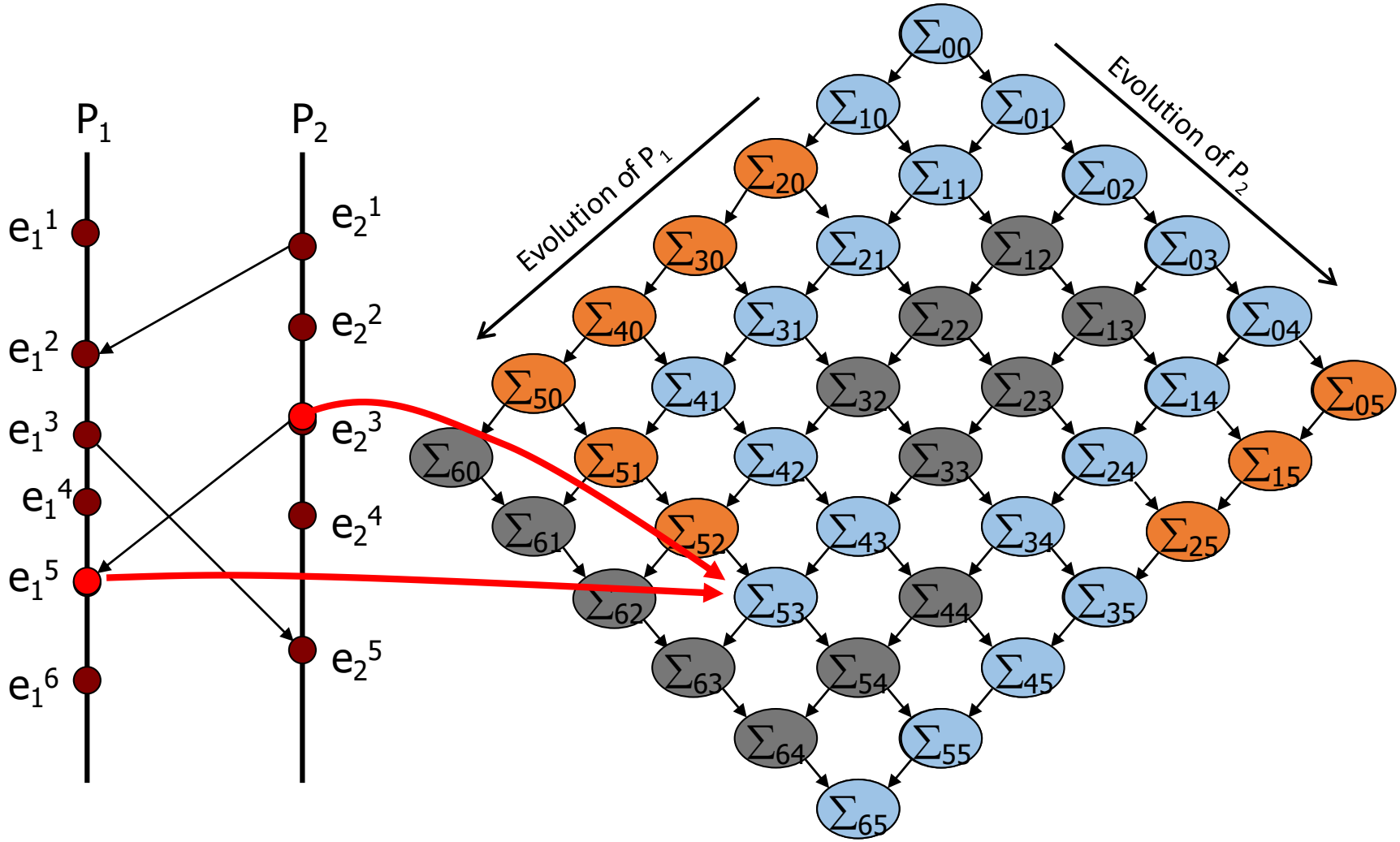
State explosion in concurrent programs



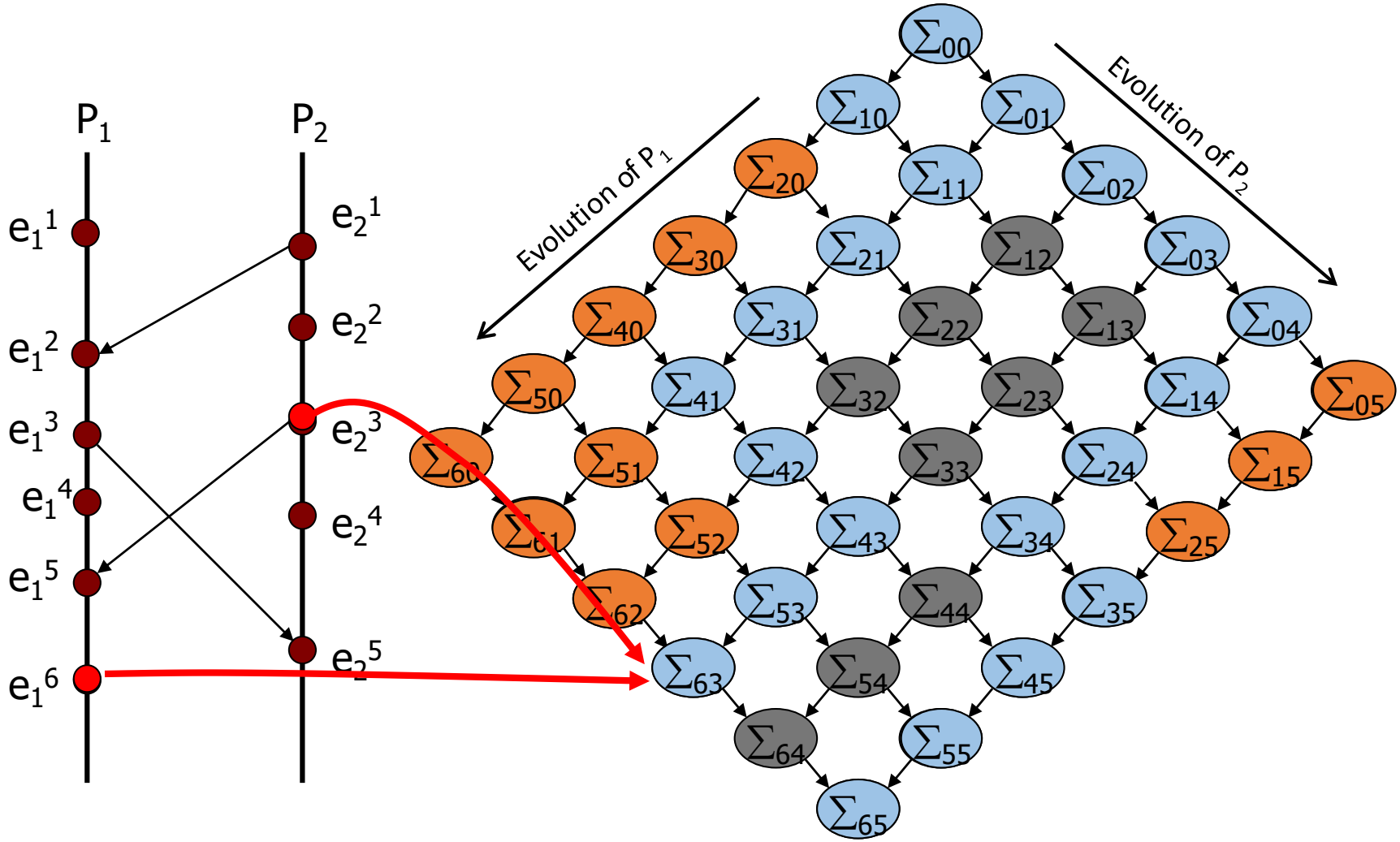
State explosion in concurrent programs



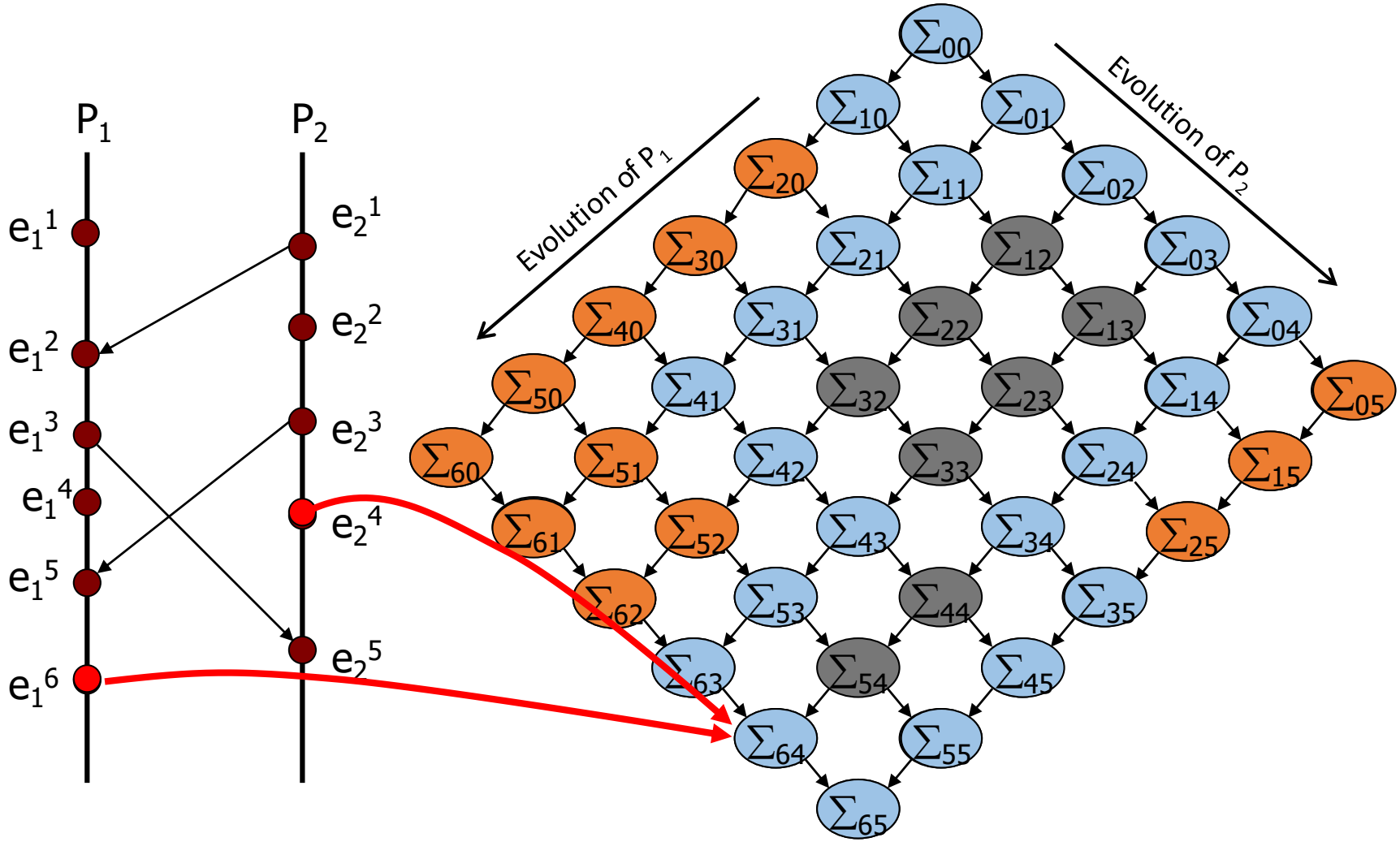
State explosion in concurrent programs



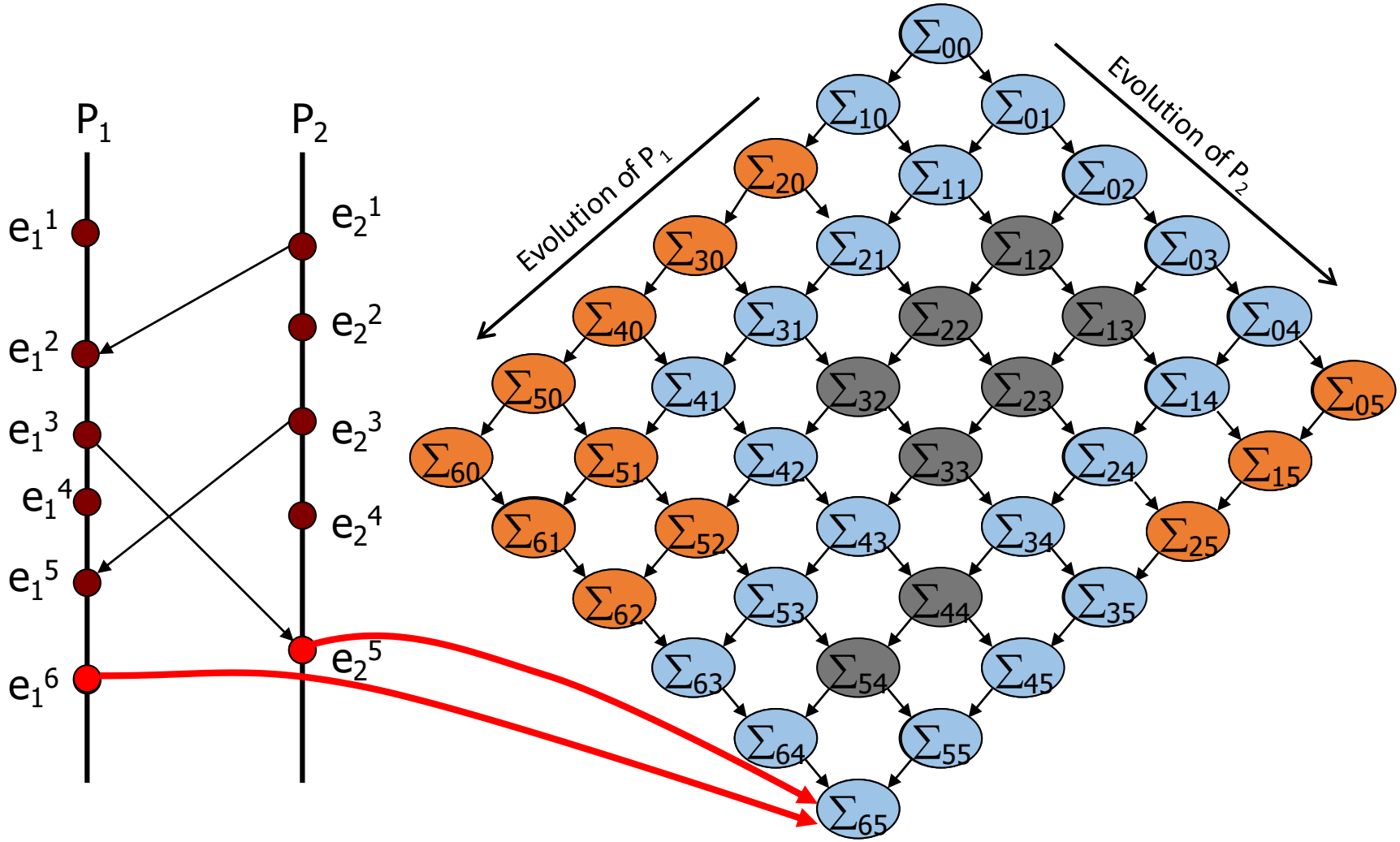
State explosion in concurrent programs



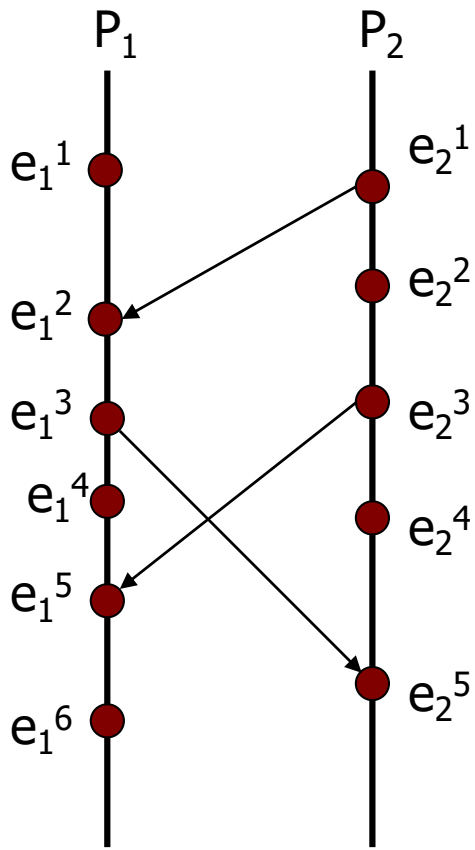
State explosion in concurrent programs

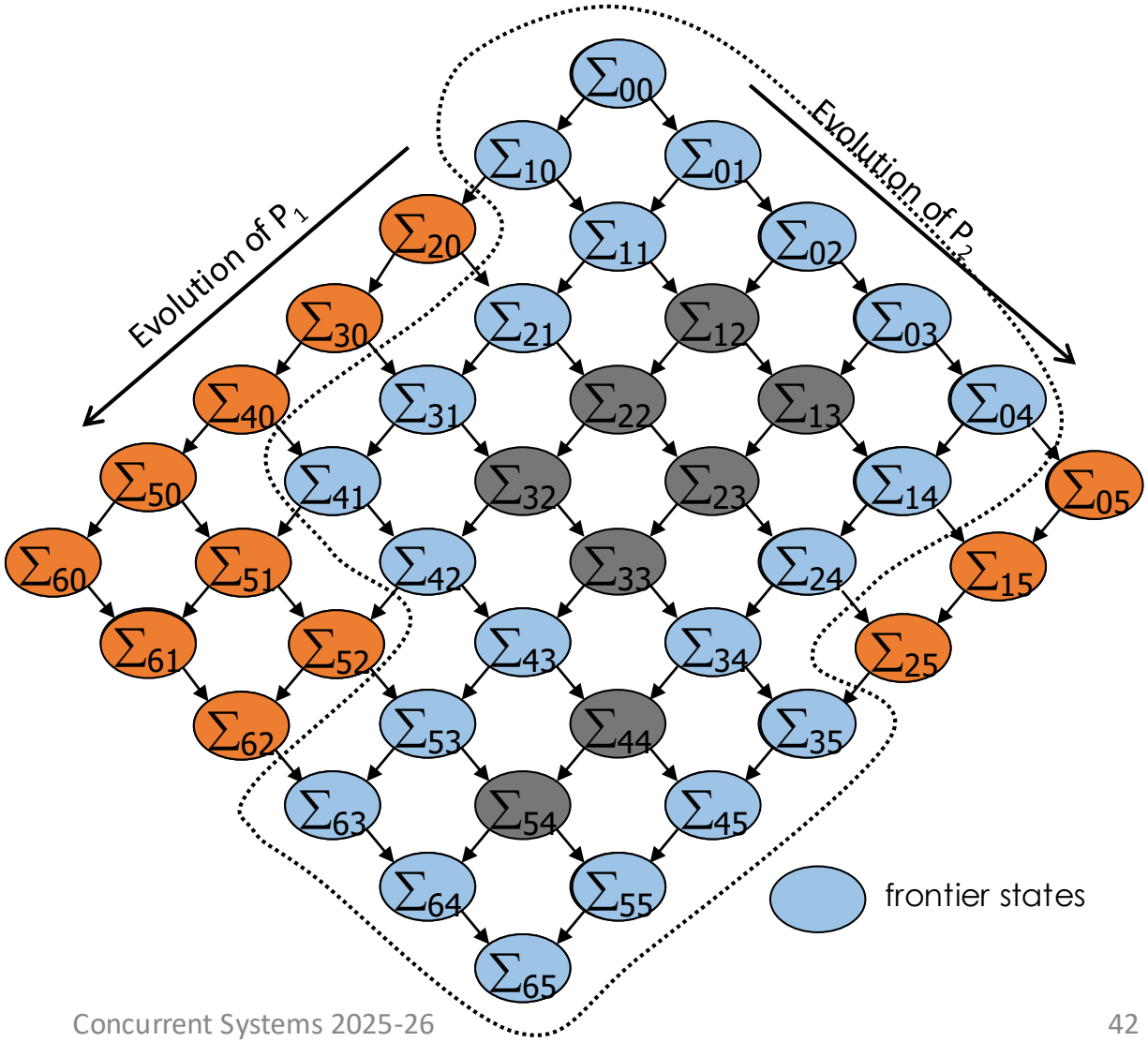


State explosion in concurrent programs

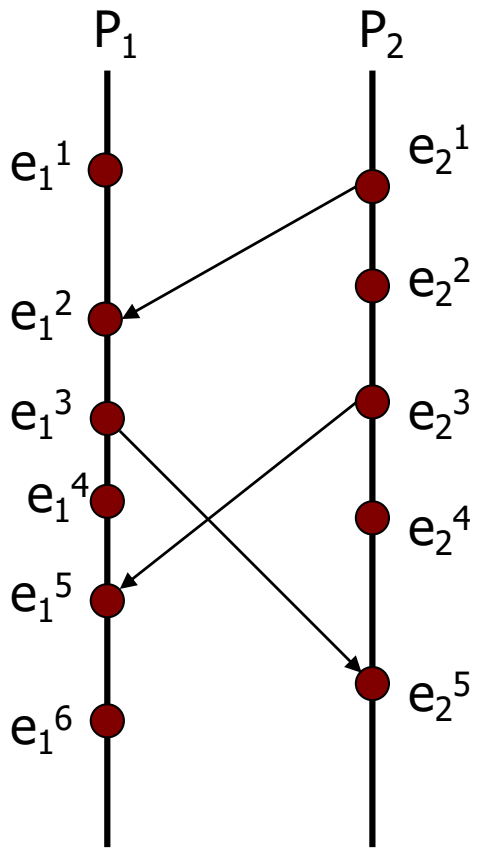


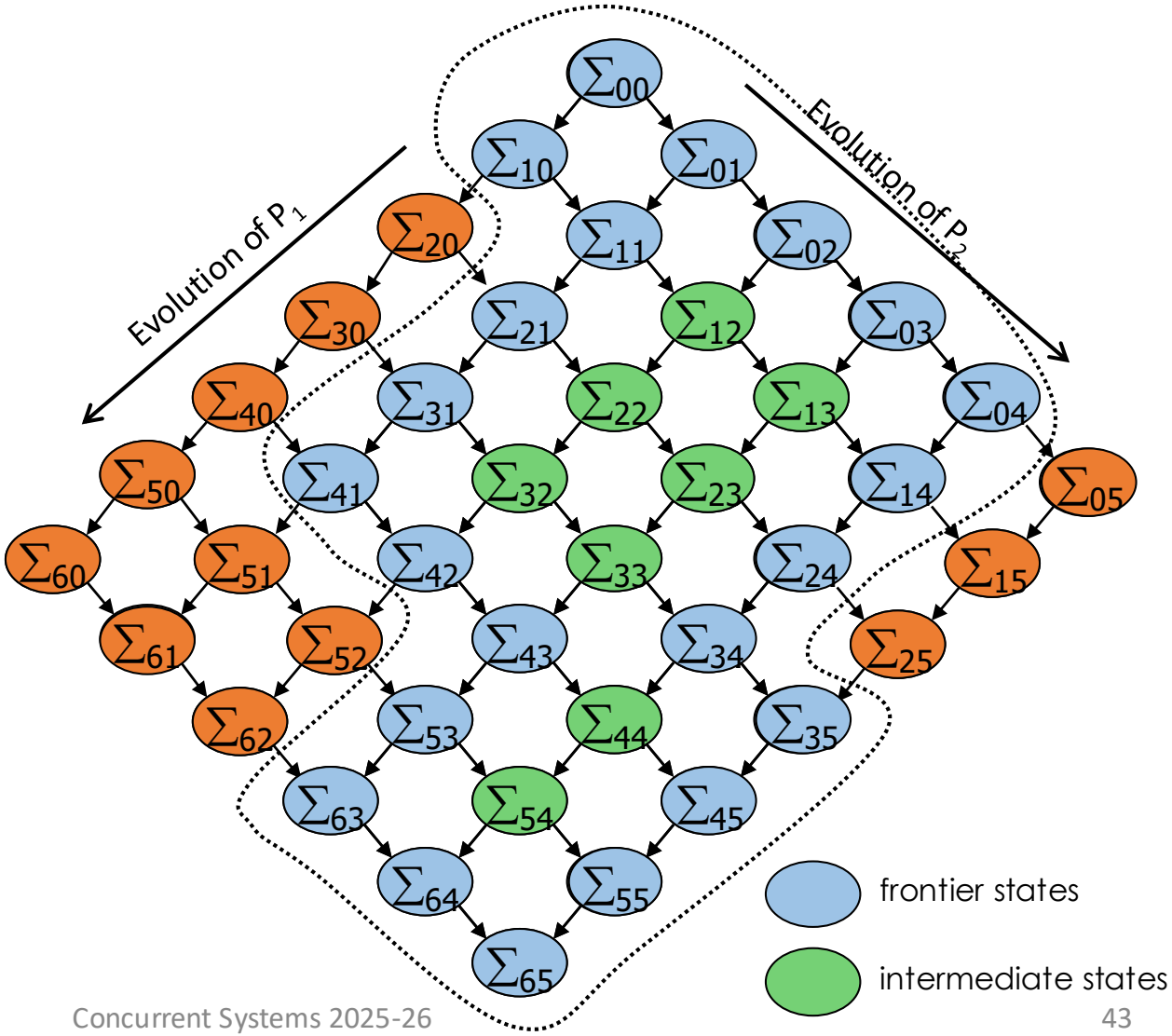
State explosion in concurrent programs



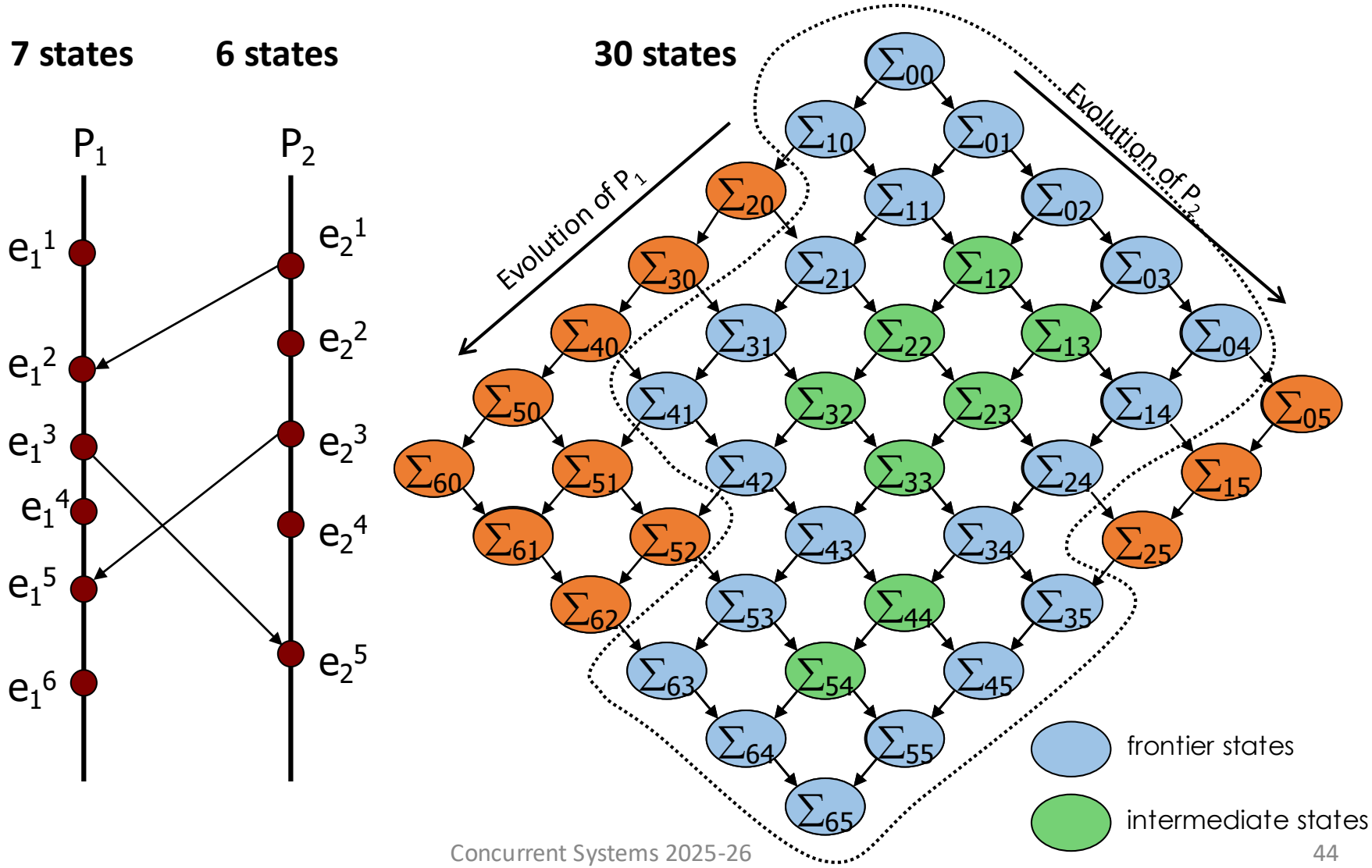


State explosion in concurrent programs

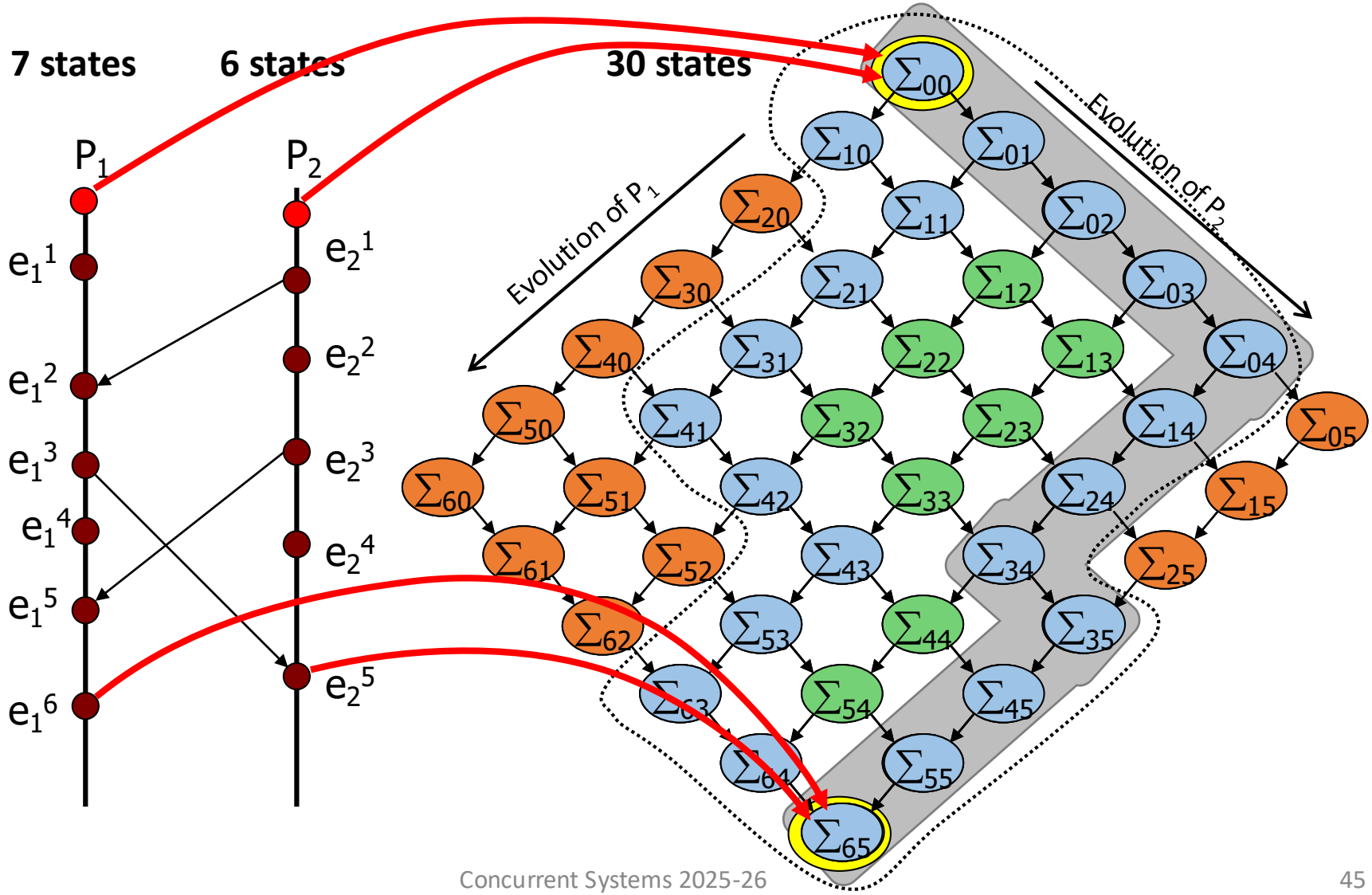




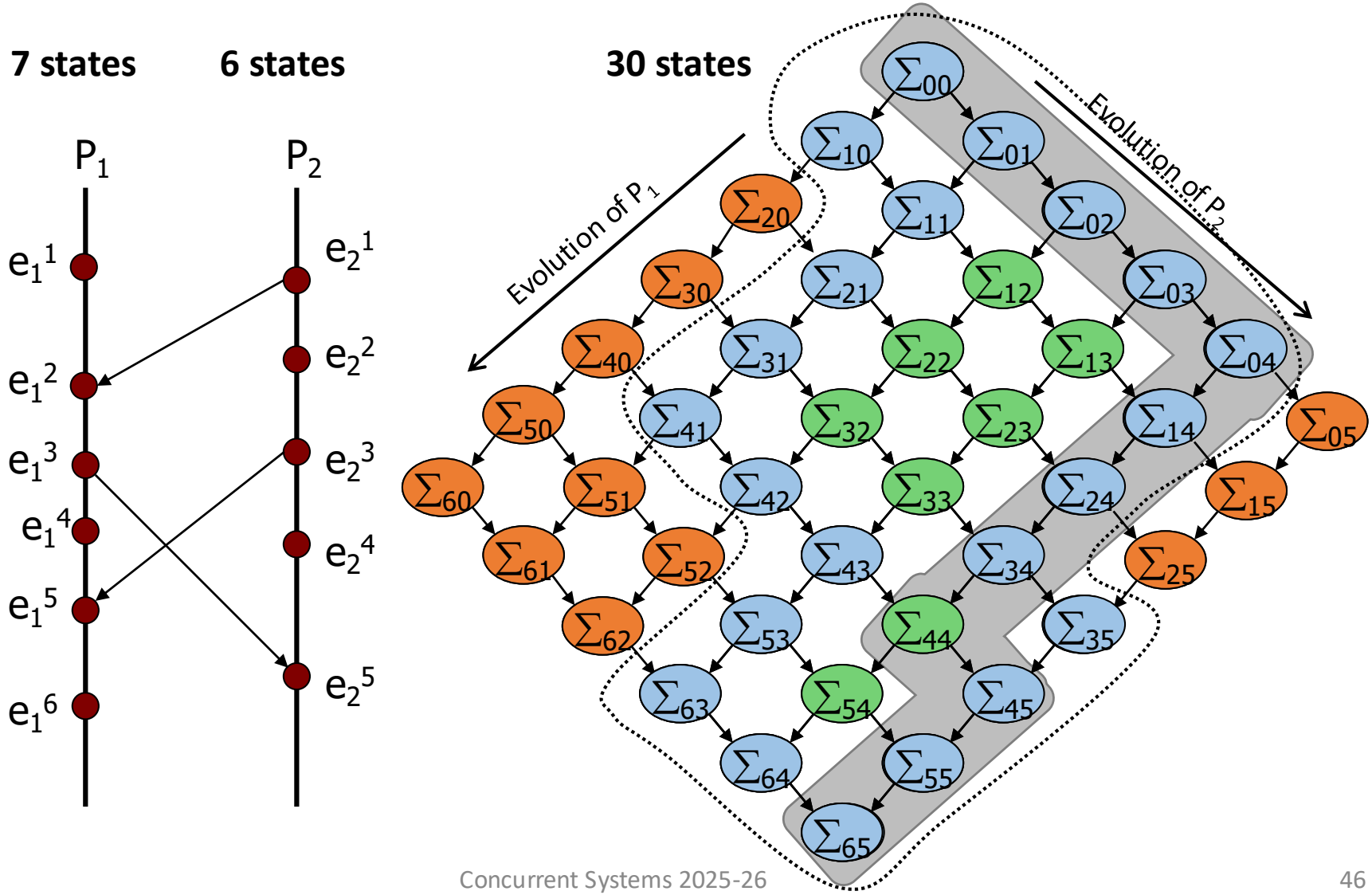
State explosion in concurrent programs



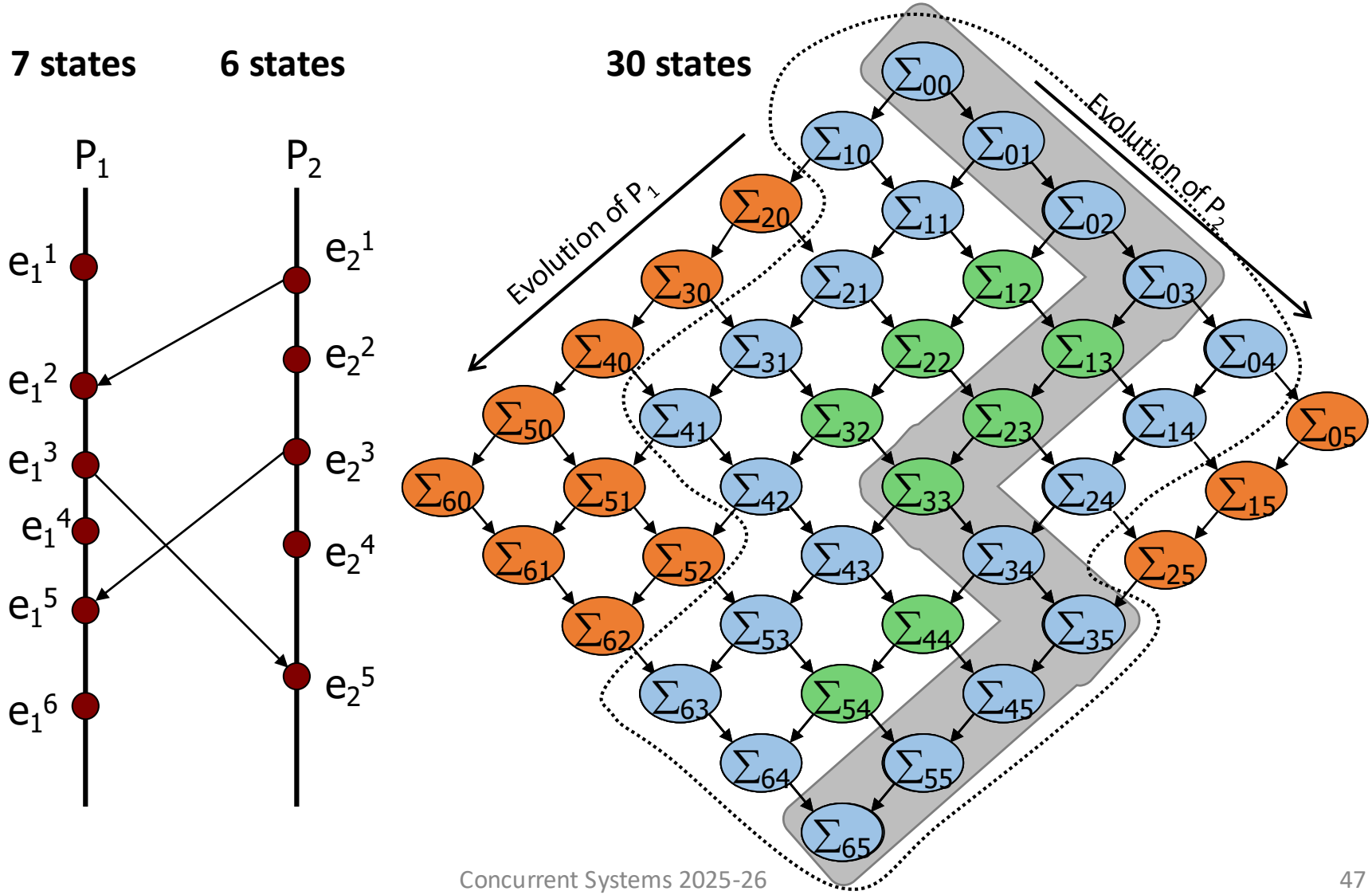
Consistent run: valid path Σ_{00} to Σ_{65}



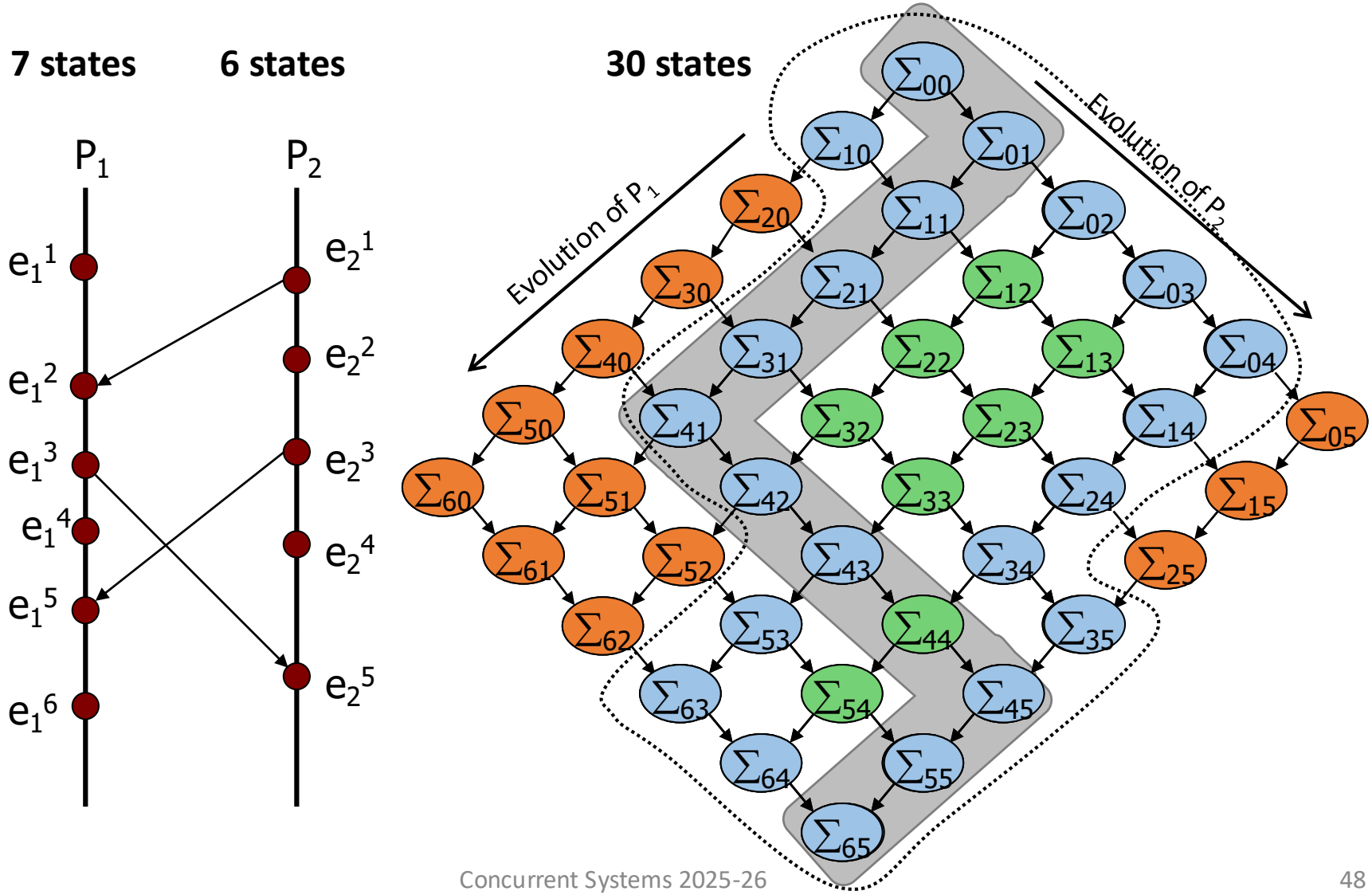
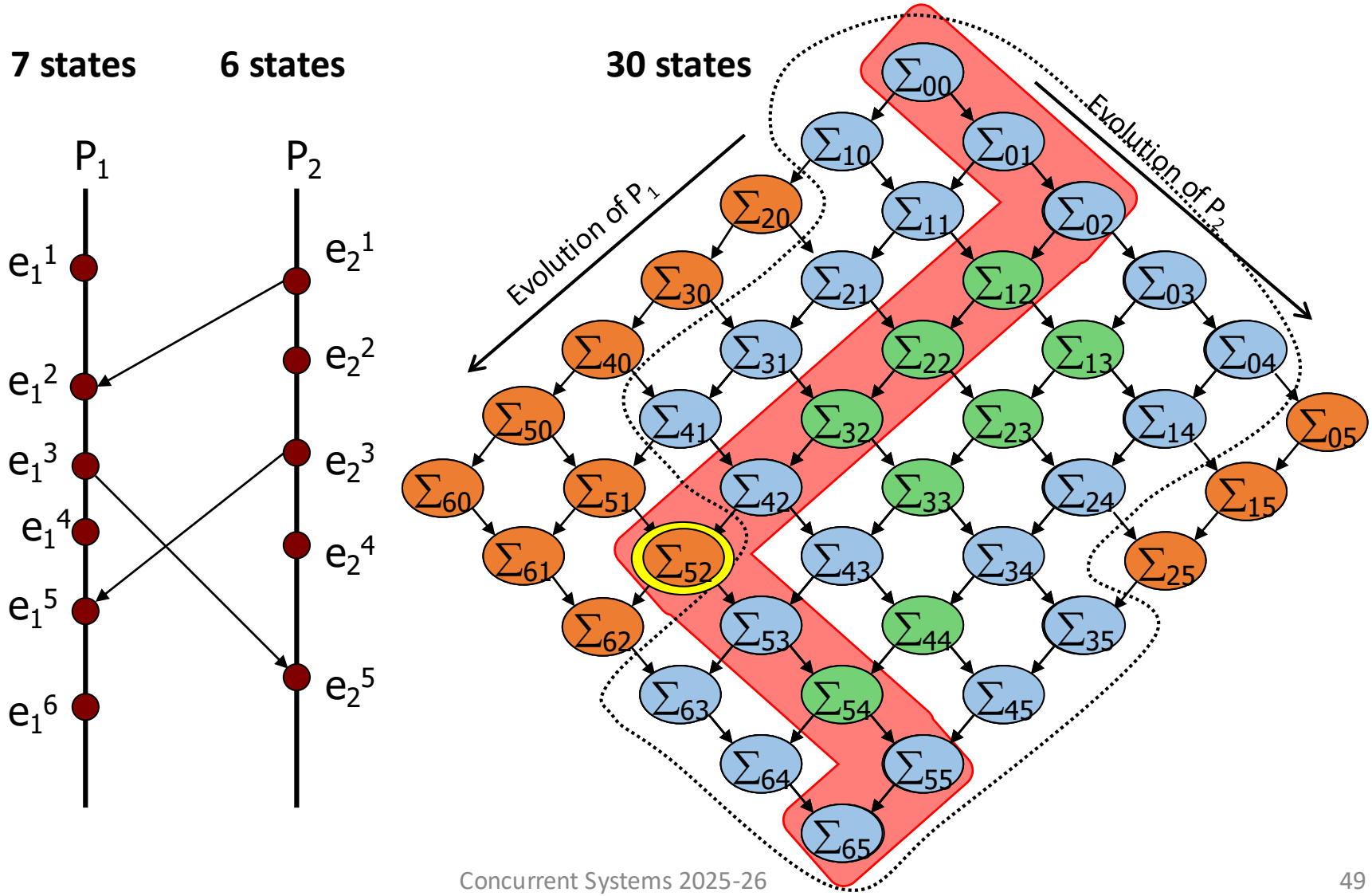
Consistent run



Consistent run



Consistent run



Inconsistent run – program error